

Econ 139: Labor Economics

Minimum Wage

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Minimum wage

- Debates around minimum wages have been raging ever since its inception in 1938
- It is both hotly debated today in politics, but is also one of the most debated topics in labor economics as well
- Still, to this day, there is no consensus on the merits or demerits of a minimum wage



Minimum Wage Debates Throughout History



1938
Fair Labor
Standards Act

1961
Coverage
Expanded

1994
Card-Krueger
Study

2025
Present

Opposing Sides

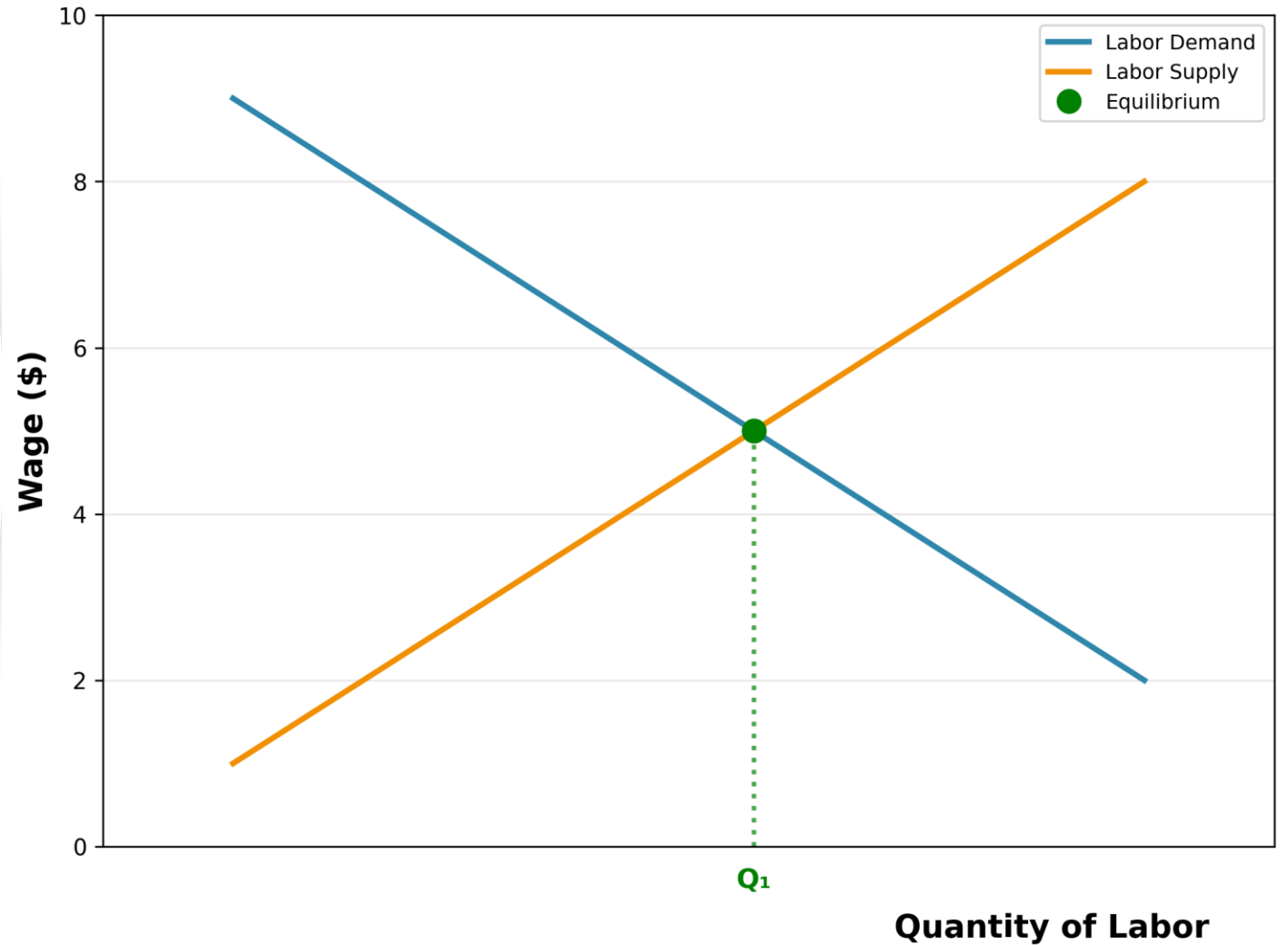
Pro Minimum Wage Argument: firms need to provide a living wage. By not having a minimum wage we are allowing the most vulnerable workers to be harmed by employers

Anti-Minimum Wage Argument: minimum wages are detrimental to vulnerable workers. If you increase the price of labor, firms will necessarily layoff workers. The very workers you are trying to help will now be unemployed

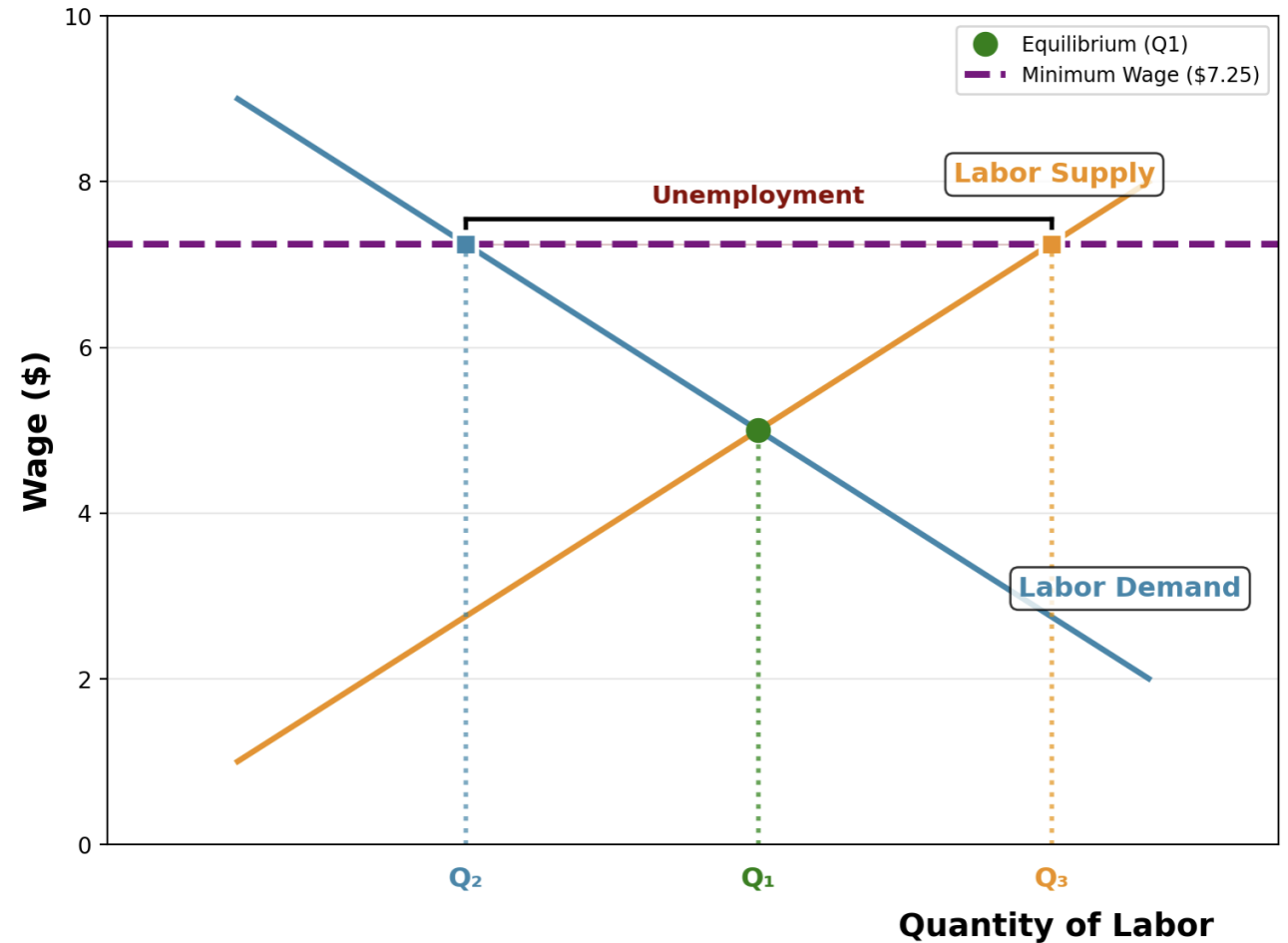
Minimum wages with Perfect Competition

- In perfectly competitive labor markets, we can model a minimum wage as a price floor
- This has very clear implications for workers and firms

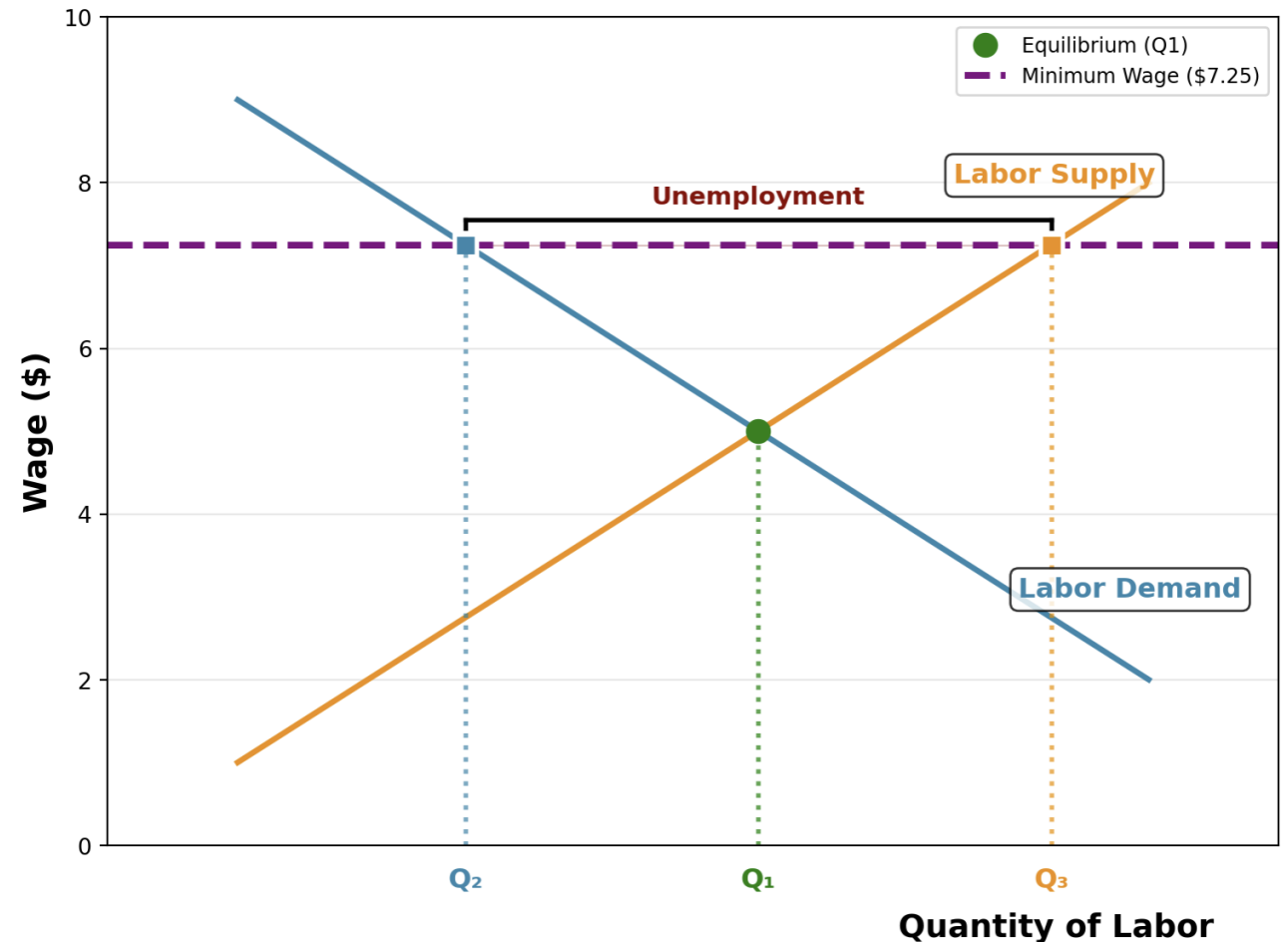
Minimum wages with Perfect Competition



Minimum wages with Perfect Competition



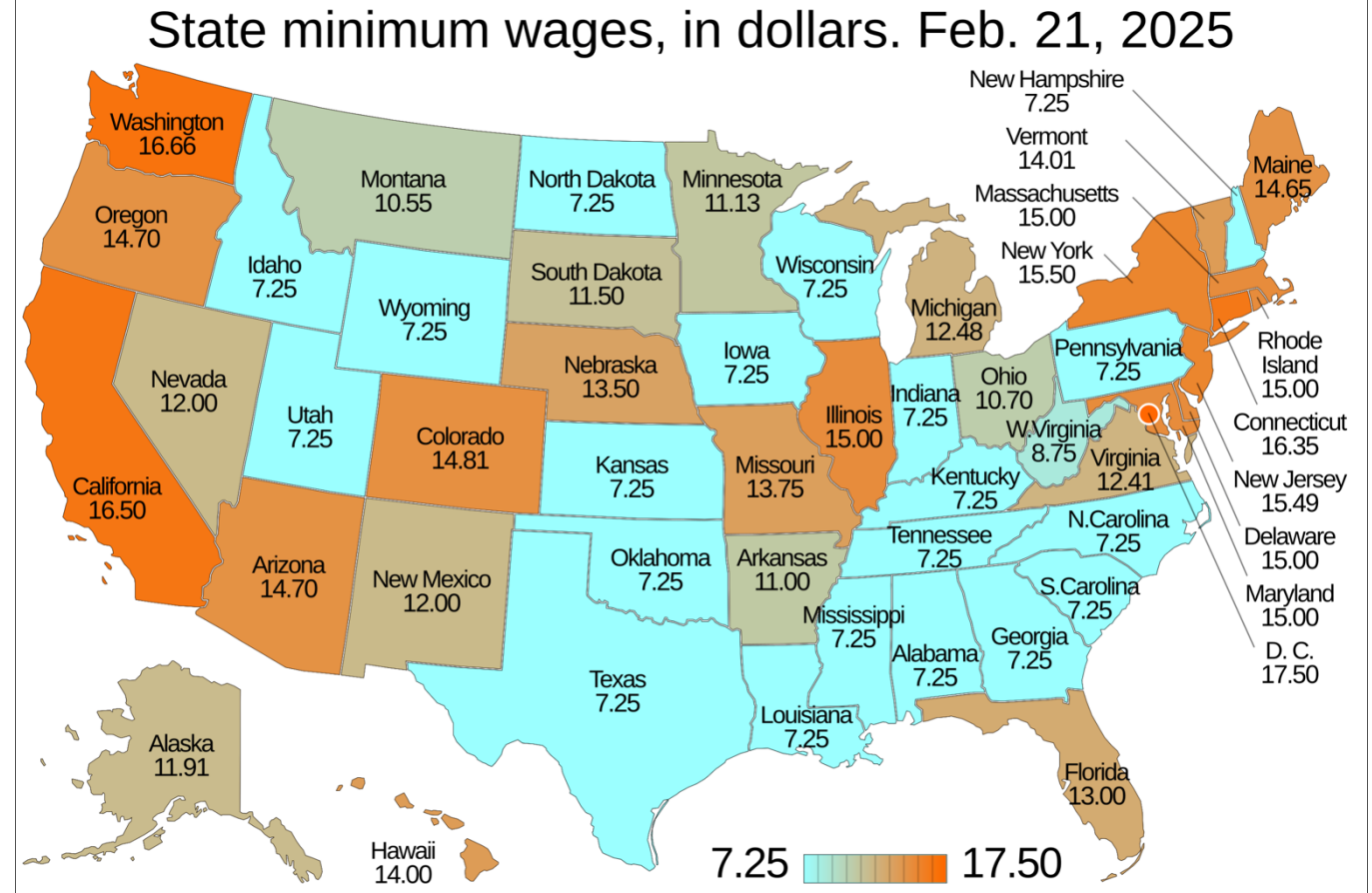
Minimum wages with Perfect Competition



Theory tells us employment will fall from Q_1 to Q_2 , but it doesn't tell us how much

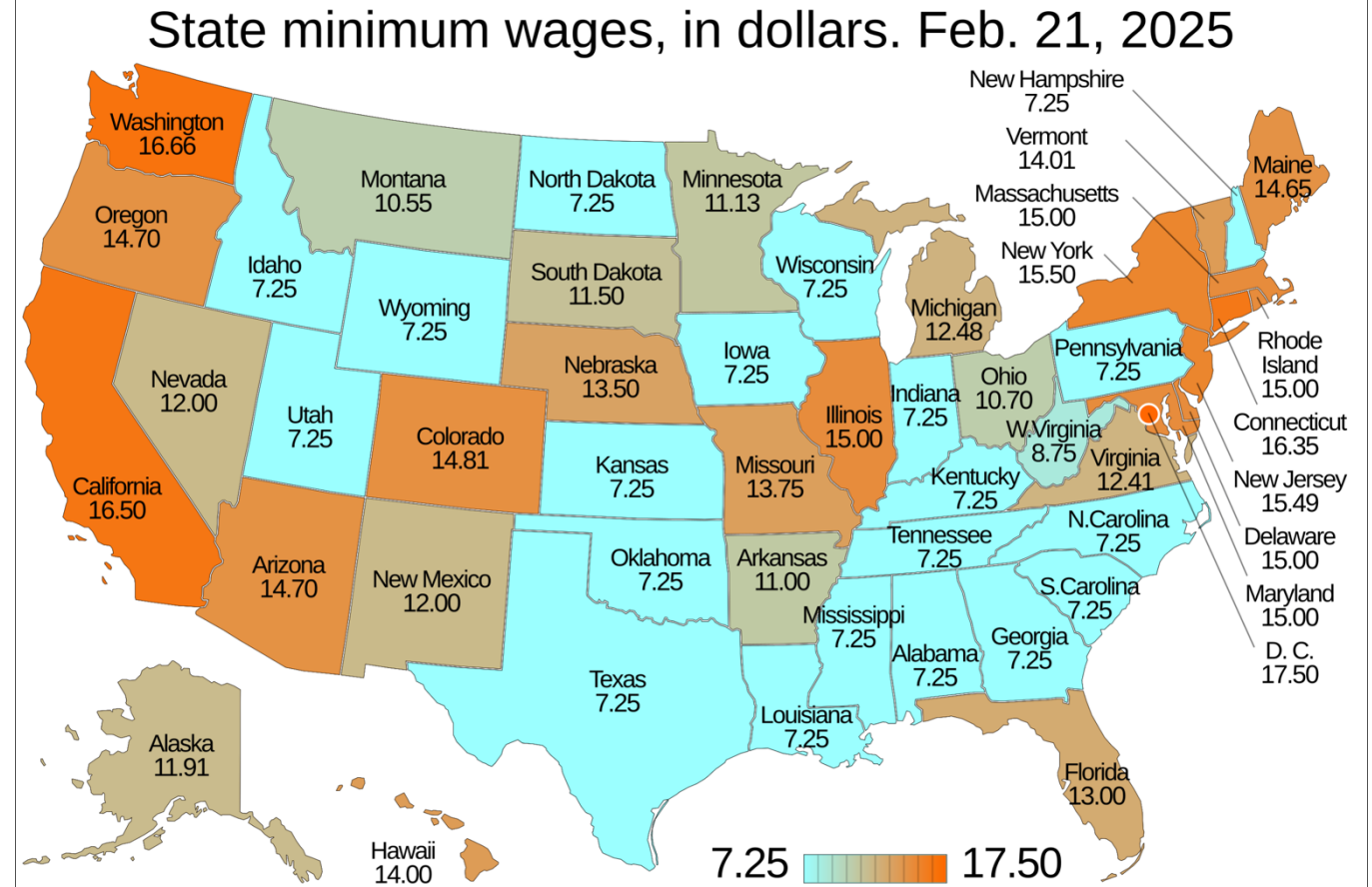
How to Estimate Impact of Minimum Wage on Employment?

- One simple idea is to use variation across states
- Washington has a very high minimum wage of \$16.66
- Alabama has no state-level minimum wage, therefore federal of \$7.25 applies



How to Estimate Impact of Minimum Wage on Employment?

- Washington and Alabama vary on a huge number of dimensions
- So if we find a difference in employment levels, maybe it is these other variables that are causing it
- Washington and Alabama have different levels of education, different costs of living, different industries, and so on...

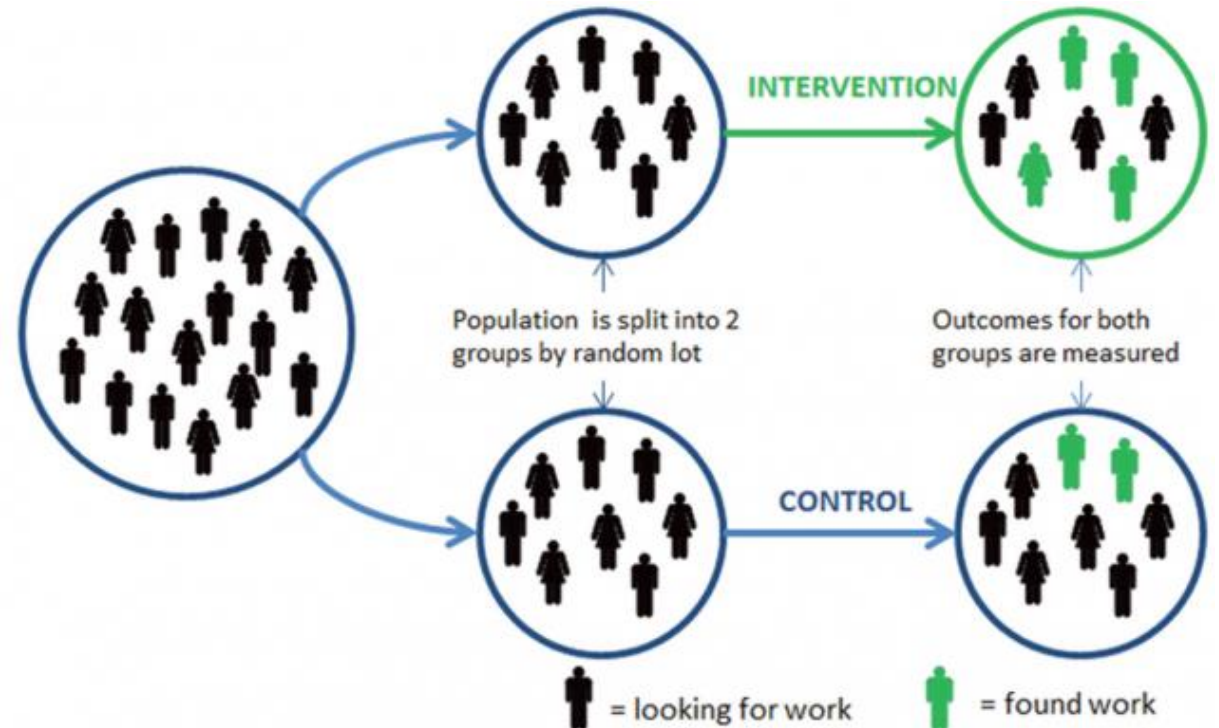


Causal Inference

- The issue of confounding variables is pervasive in economics
- There are many situations in which we want to understand the causal effect of some variable X (e.g. minimum wage laws) on some variable Y (e.g. employment)
- A **confounding factor** is anything that may be associated with both X and Y
- The idea of confounding factors is extremely important, not only in economics, but science more broadly defined

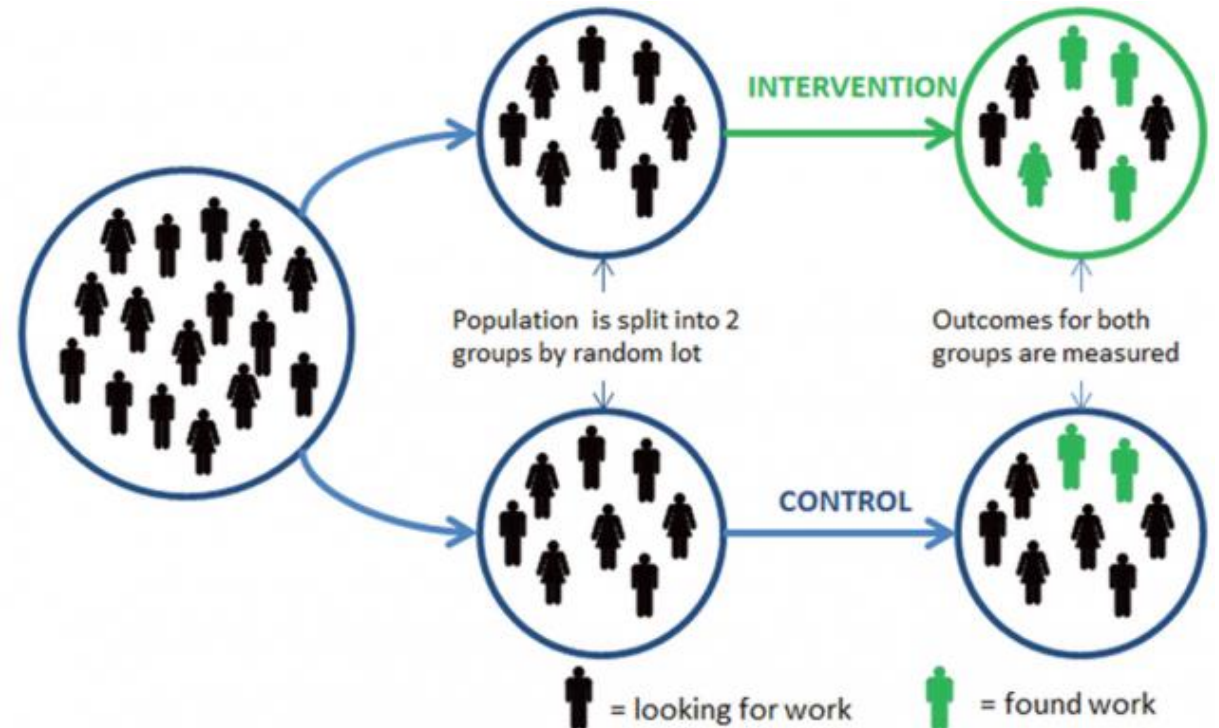
Experiments

- In medicine, there is an established way to make causal claims: The Randomized Control Trial (RCT)
- Randomize who receives treatment vs. control. Then any differences between groups must be due to the treatment



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- In medicine, there is an established way to make causal claims: The Randomized Control Trial (RCT)
- Randomize who receives treatment vs. control. Then any differences between groups must be due to the treatment
- Random assignment assures there are not other differences across the groups on average



Natural Experiments

- In Economics (and other social sciences), it often isn't possible to run a Randomized Control Trial
- Therefore, a popular alternative is to find **natural experiments**
 - These are situations that mimic an experiment. Some people, firms or regions received a treatment of interest, while others did not
- Let's discuss trying to find a natural experiment for the minimum wage

A Natural Experiment for the Minimum Wage

- In 1992, New Jersey increased its minimum wage from \$4.25 to \$5.05
- Our goal in this next section is to understand the impact of this change on employment
- To focus our discussion, we will consider the fast-food industry
 - Seems like a natural focus given fast food workers are often paid minimum wage.

Change in Employment in New Jersey

- One simple thing to do would be to measure employment in fast food restaurants in 1992 and compare to what it was in 1991

$$\Delta E_{NJ} = E_{NJ, 1992} - E_{NJ, 1991}$$

- For example, if fast food restaurants on average employed 20 workers in 1991, but 18 in 1992, then $\Delta E_{NJ} = -2$

Change in Employment in New Jersey

- Can we simply use ΔE_{NJ} as our estimate of the employment effect of the minimum wage
- No! Why?
 - Again, we have some confounding factors. There are fluctuations in employment across time that have nothing to do with minimum wage laws

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Counterfactual

- What we need is a **counterfactual** for New Jersey
 - A **counterfactual** tells us what would have happened to New Jersey employment had the minimum wage not changed
- It is extremely difficult to find a good counterfactual (this is why randomized control trials are so useful)
- But in this case, why don't we see if nearby Pennsylvania is a good counterfactual

Counterfactual



Counterfactual

Lambertville, NJ



Yardley, PA



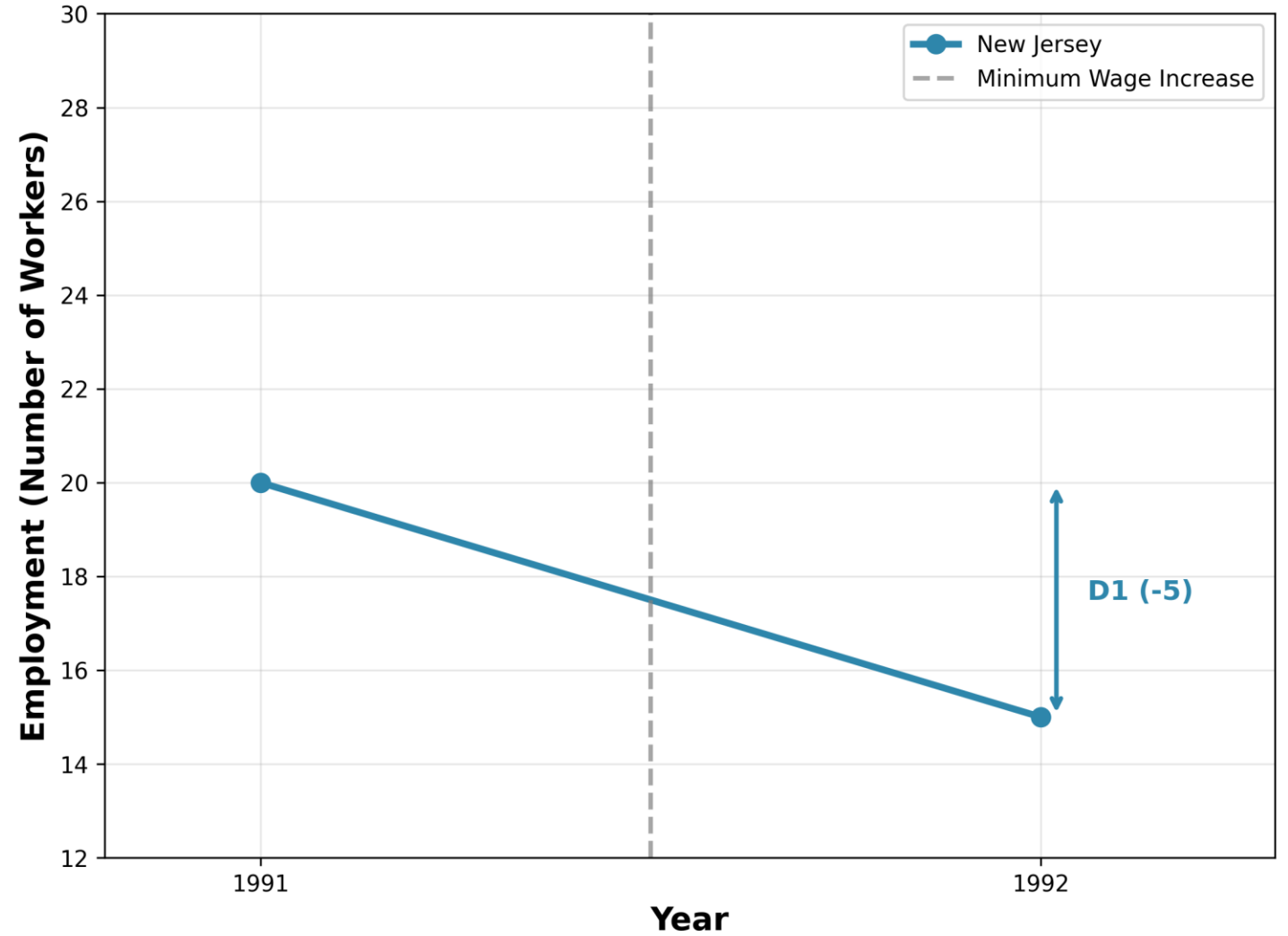
Counterfactual

- Let $\Delta E_{PA} = E_{PA,1992} - E_{PA,1991}$
- This is what we will use as the counterfactual for NJ
- The treatment effect is the difference between the actual change in employment in NJ and the predicted based on changes in PA. Lets call this treatment effect ΔE

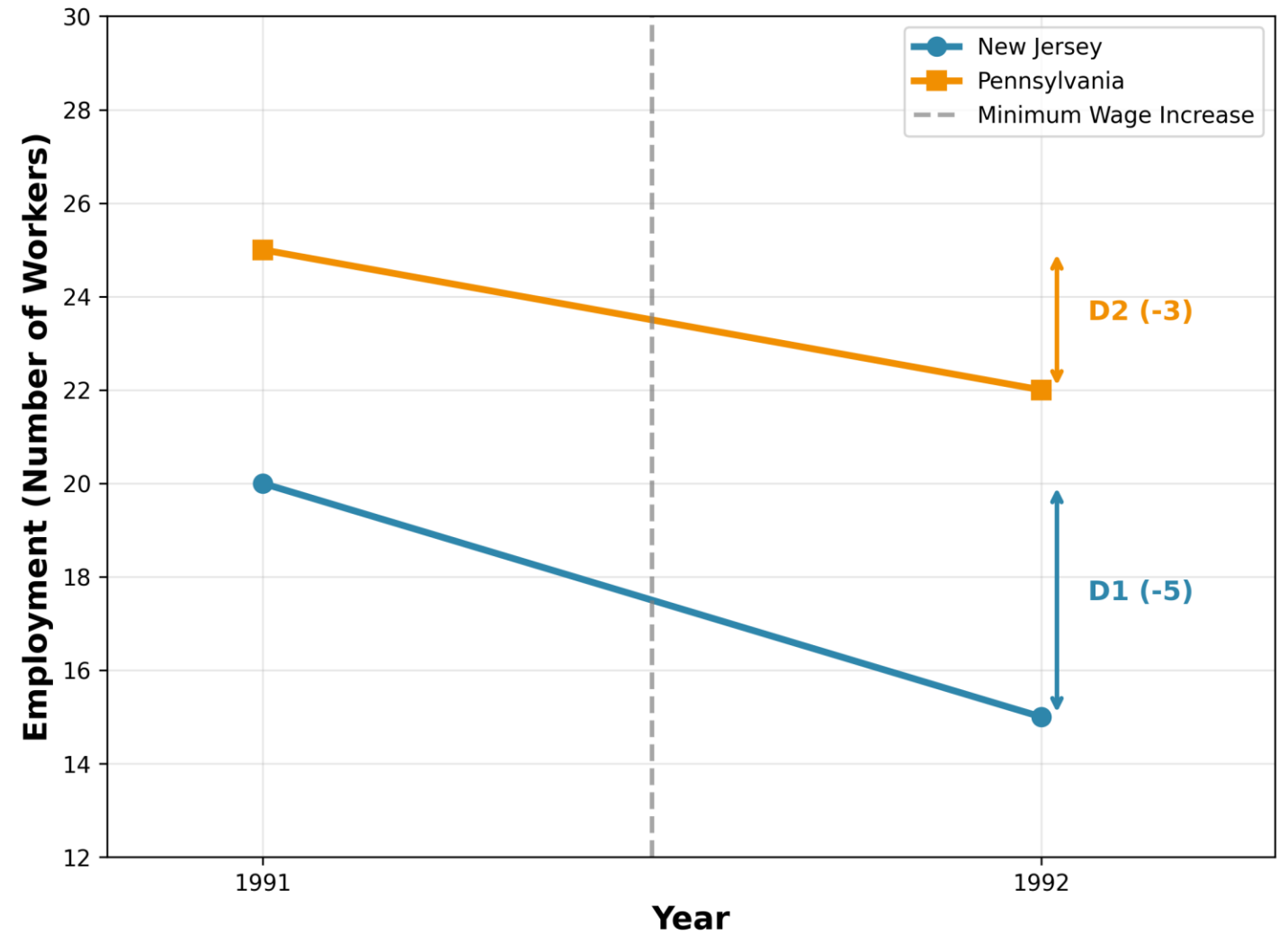
$$\Delta E = \Delta E_{NJ} - \Delta E_{PA}$$

- This is referred to as a difference-in-differences estimator

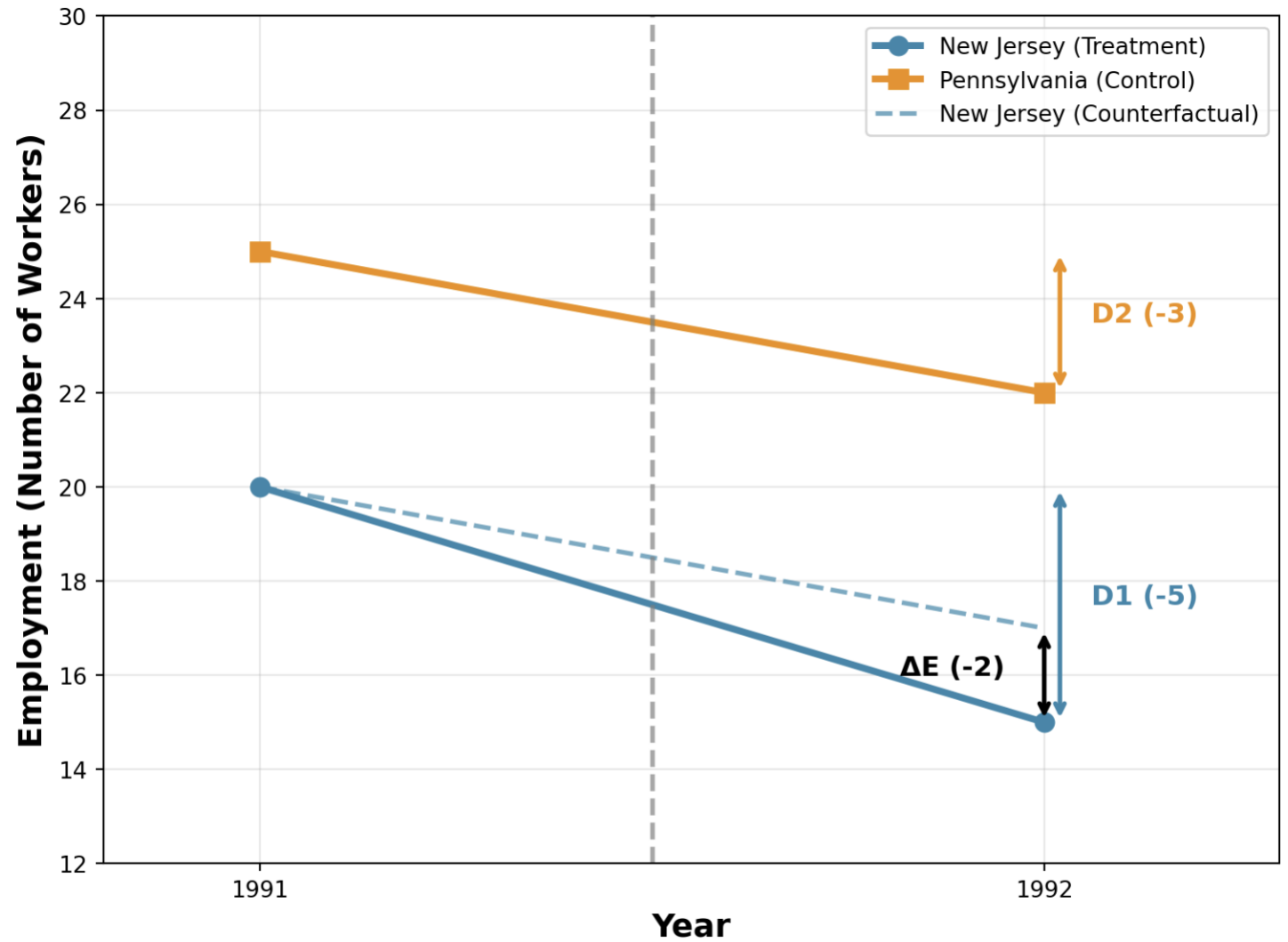
Difference-in-differences visualization



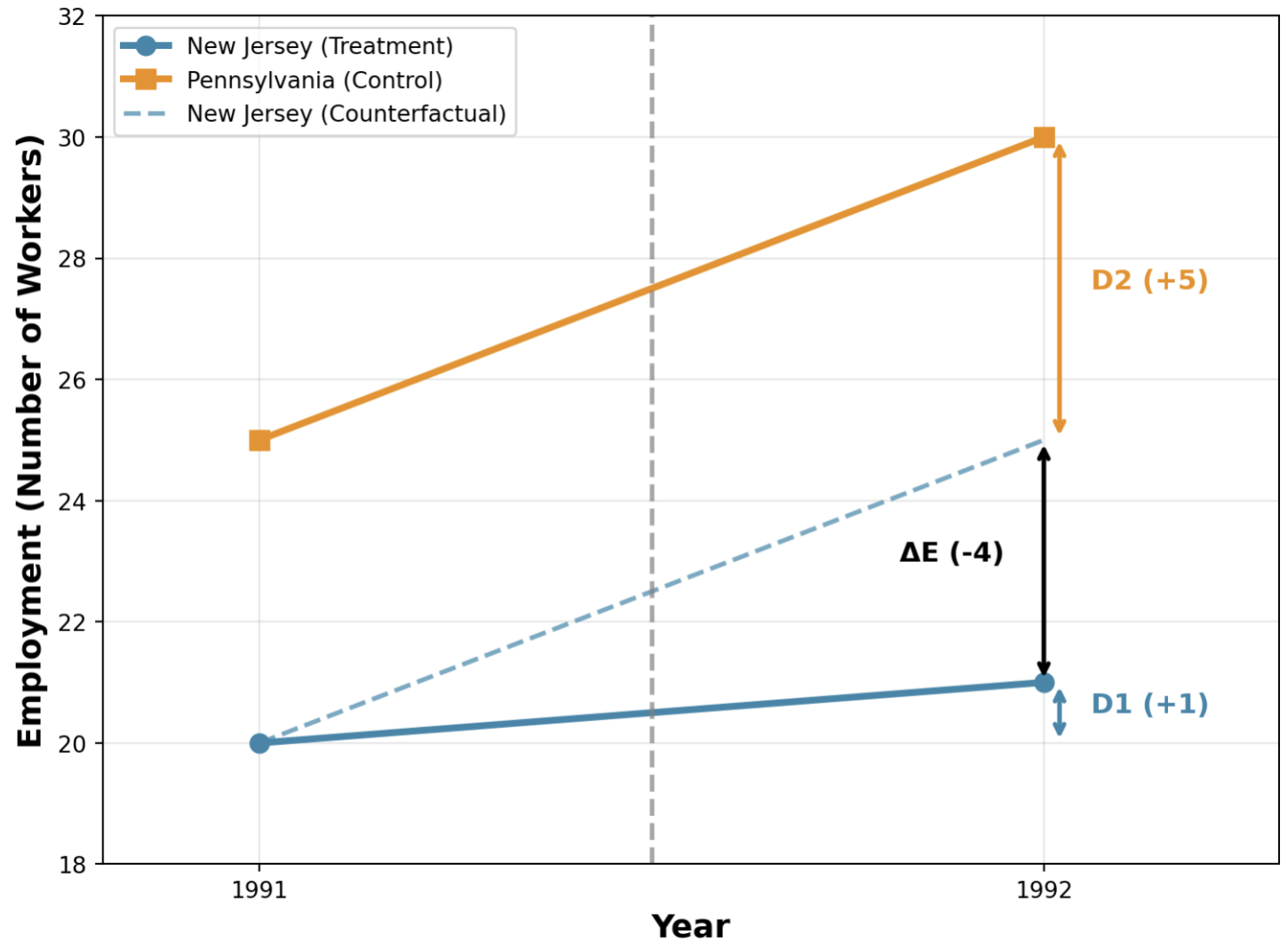
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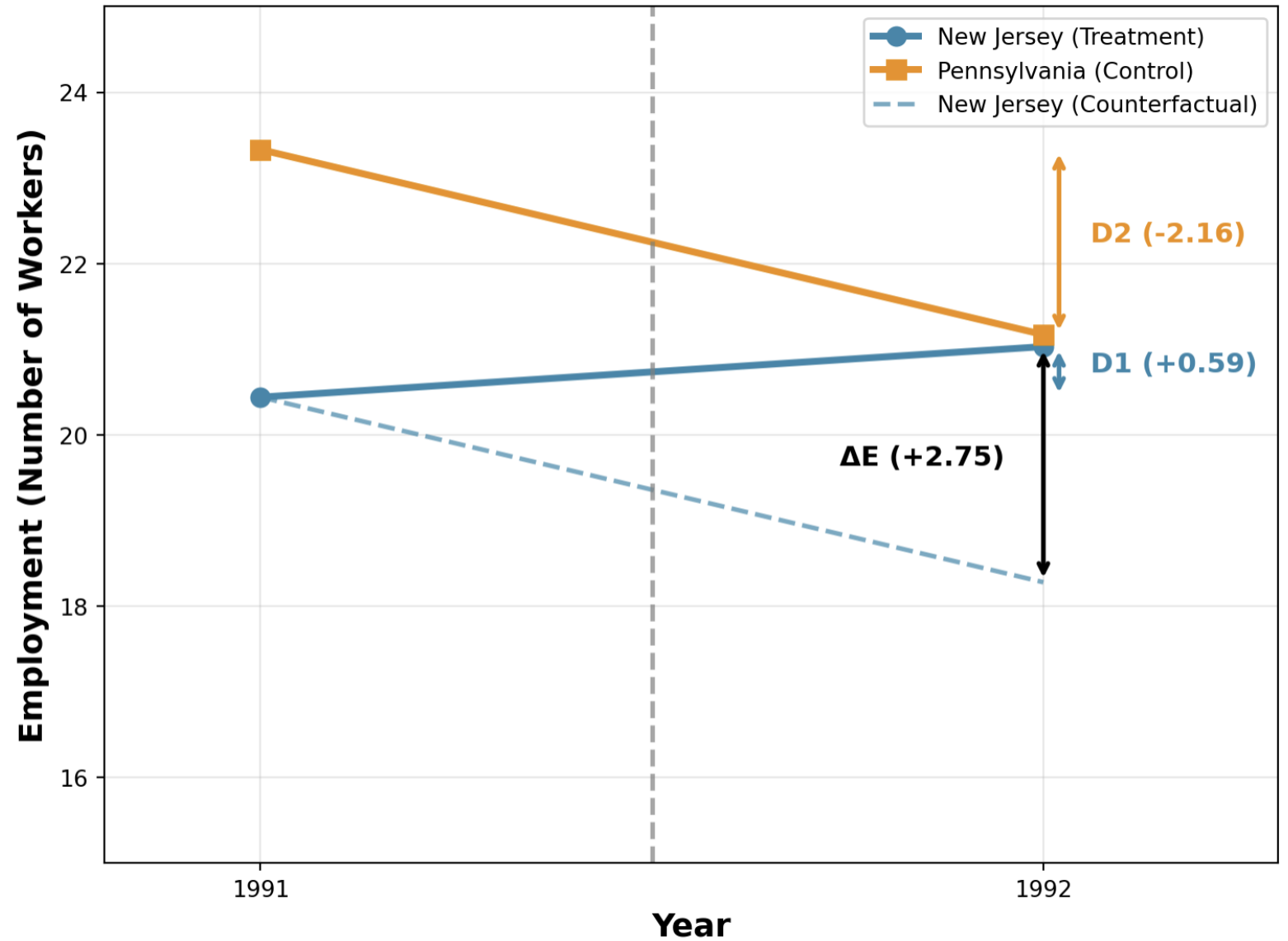
Card and Krueger (1994)

- Now let's go back to Card and Krueger's study
- Contacted many stores by phone and had very high response rates

TABLE 1—SAMPLE DESIGN AND RESPONSE RATES

		Stores in:	
	All	NJ	PA
<i>Wave 1, February 15–March 4, 1992:</i>			
Number of stores in sample frame: ^a	473	364	109
Number of refusals:	63	33	30
Number interviewed:	410	331	79
Response rate (percentage):	86.7	90.9	72.5
<i>Wave 2, November 5–December 31, 1992:</i>			
Number of stores in sample frame:	410	331	79
Number closed:	6	5	1
Number under renovation:	2	2	0
Number temporarily closed: ^b	2	2	0
Number of refusals:	1	1	0
Number interviewed: ^c	399	321	78

Card and Krueger Results



Card and Krueger (1994)

- Let's take a moment to appreciate how surprising this result is
 - You increase the cost of labor and *more* workers are hired
- Card and Krueger (1994) was an extremely influential paper for two reasons
 - This result is very surprising. A large number of papers have analyzed whether it holds up with alternative designs/settings
 - Was one of the first uses of a natural experiment that utilizes difference-in-differences

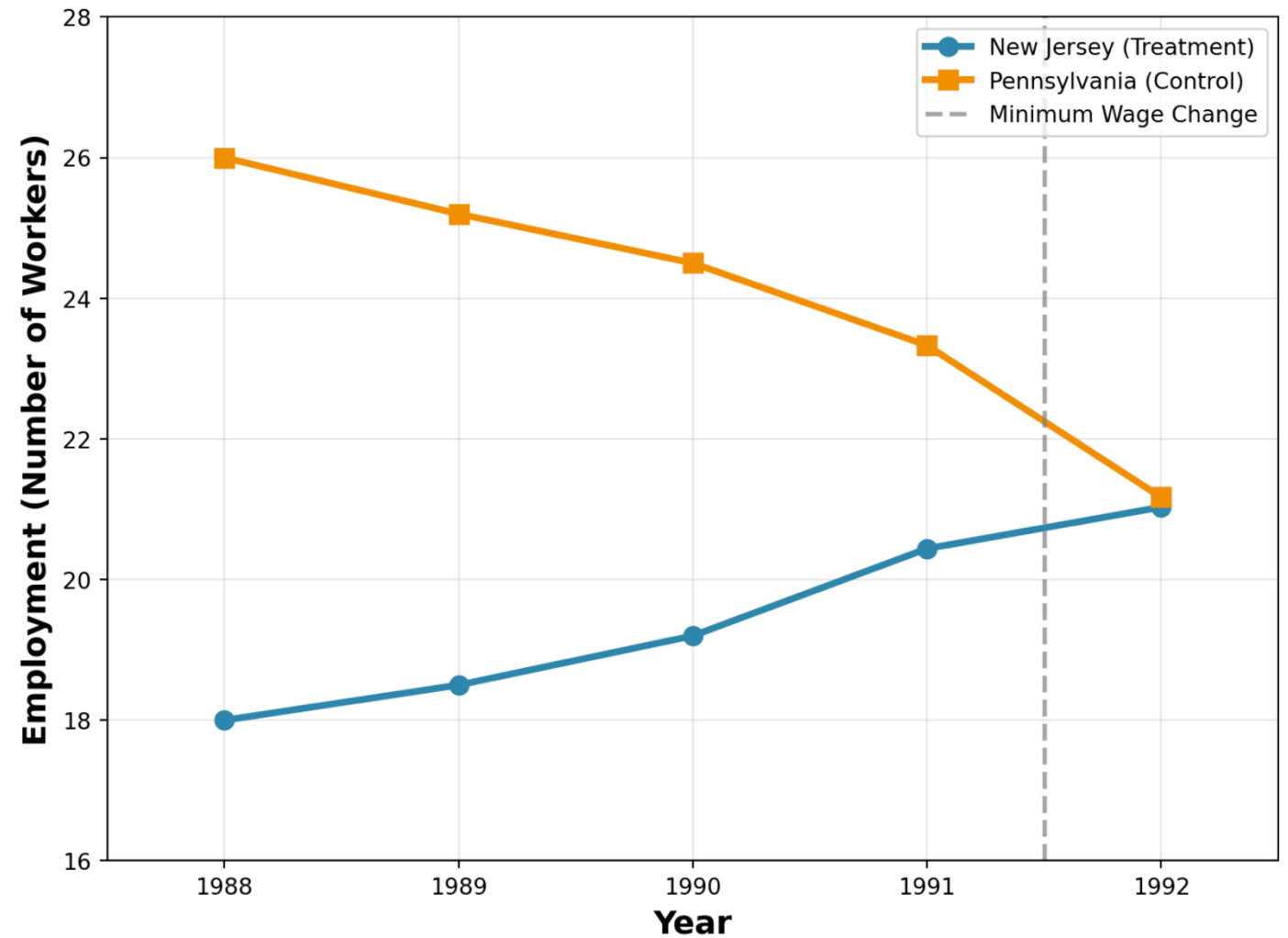
Critiques

- One natural critique is that our assumption that PA is a good counterfactual for NJ is just incorrect
 - A lot of empirical papers main focus is arguing they have found a valid counterfactual, a difficult problem in many settings
- Since Card and Krueger (1994), studying outcomes *prior* to the event of interest has become standard

Parallel Trends

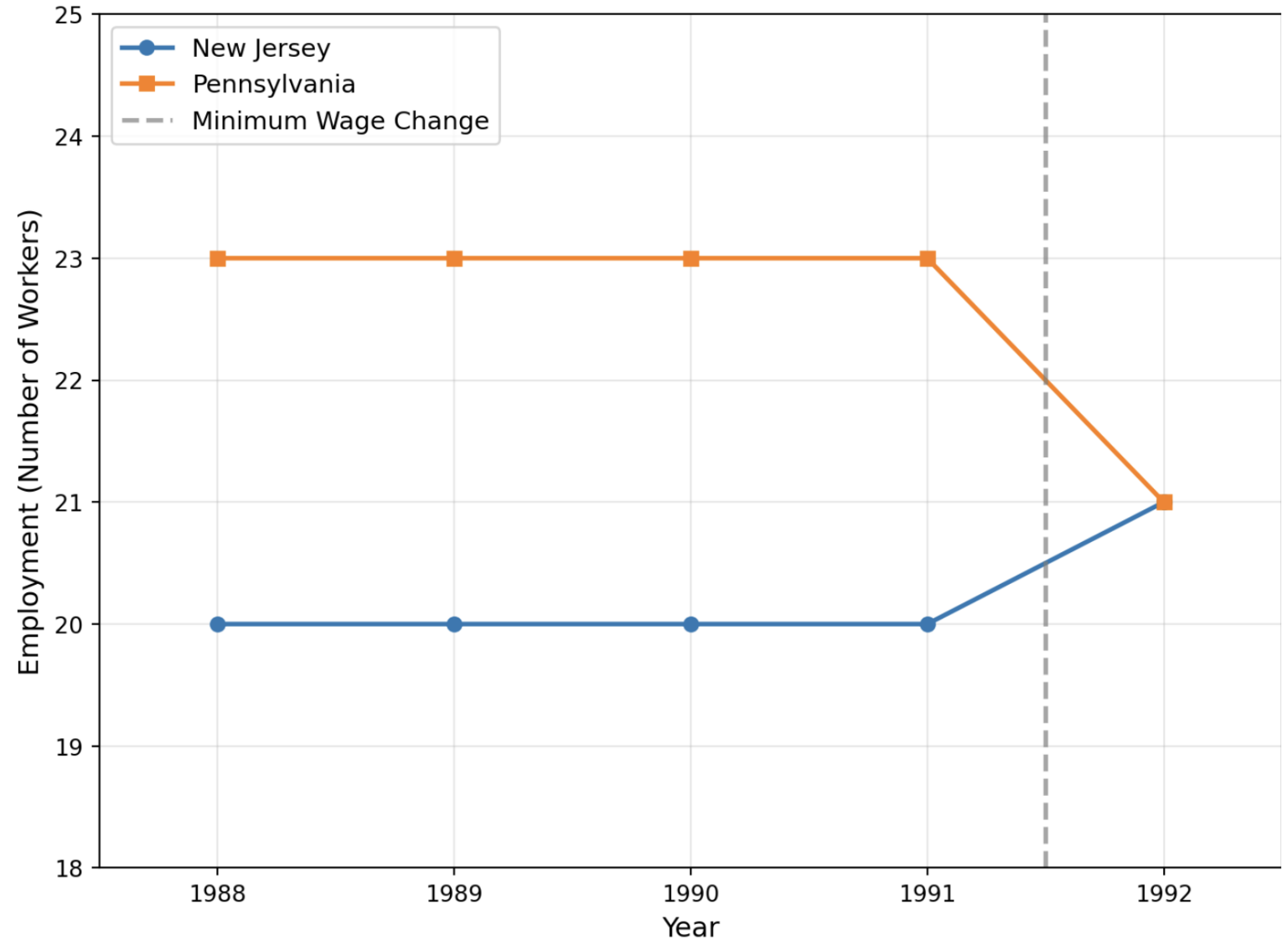
- Basic idea: if PA is a good counterfactual for NJ, then in the years before the minimum wage, their employment trajectories should be similar
- If PA is not a good counterfactual, then employment will already be trending differently prior to the minimum wage change

Hypothetical Scenario - Parallel Trends Violated



Hypothetical Scenario - Parallel Trends Validated

For modern studies
showing pre-trends is
essential



Modern Studies

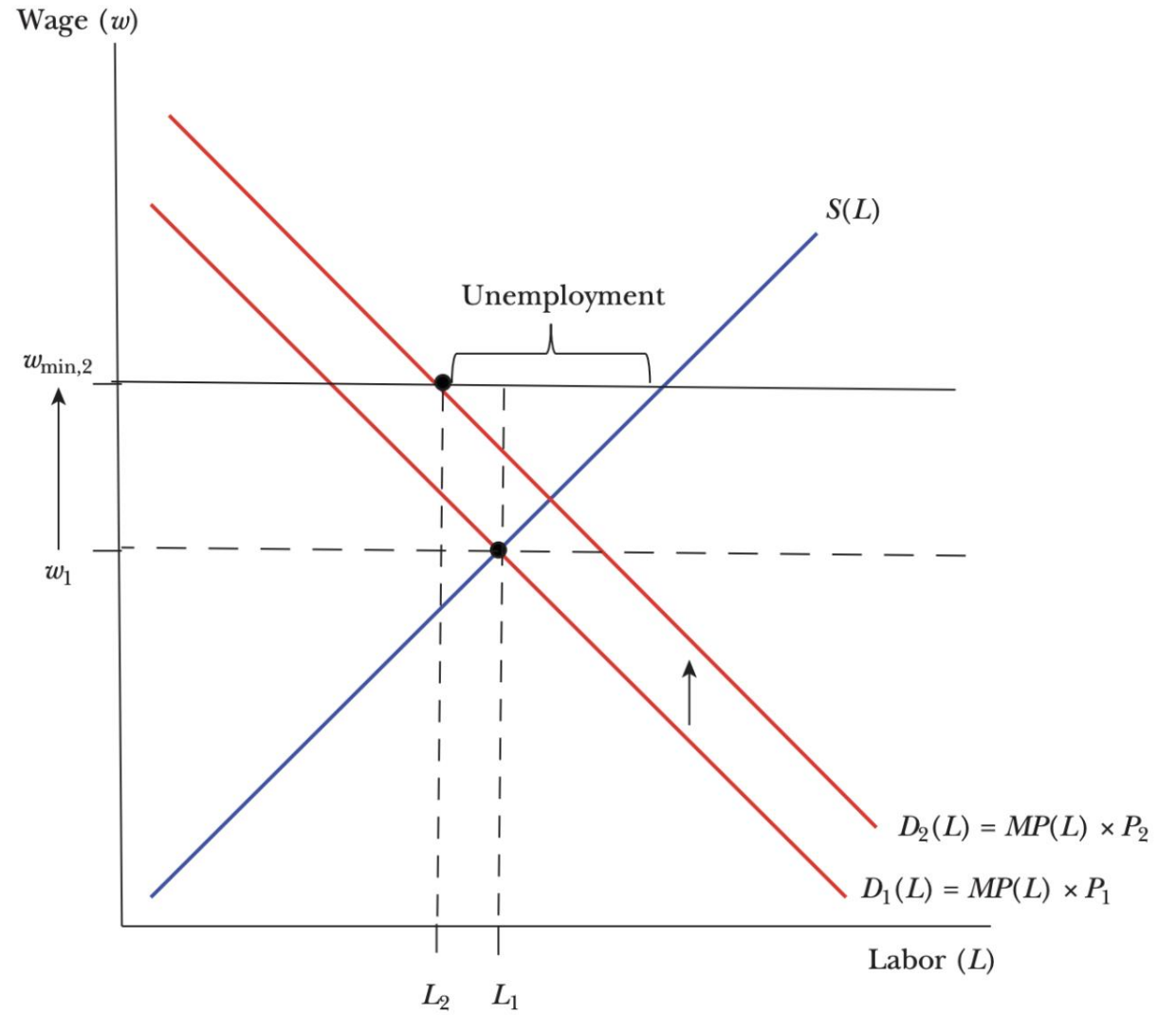
- Since the publishing of Card and Krueger (1994), hundreds of studies on employment effects of minimum wage studies have been published
 - Neumark and Shirley (2022) provide a meta-analysis and argue 80 percent of papers find **negative** employment effects while 20 percent find either no or positive effects
- Even among meta-studies, there is some disagreement. Wolfson and Belman (2019) report that they find “no support for the proposition that the minimum wage has had an important effect on U.S. employment”

My Evaluation of the Minimum Wage literature

- Even today, you will find different opinions among academic economists
- My view reading this literature is that relatively small minimum wage changes have very little impact on overall employment
 - Most minimum wage changes are relatively small, not on the scale of the Fight for 15 movement which would double the minimum wage in many place. We just don't have much evidence on changes like this
- However, we have only studied one margin of adjustment: employment. It is possible firms respond in a number of different ways

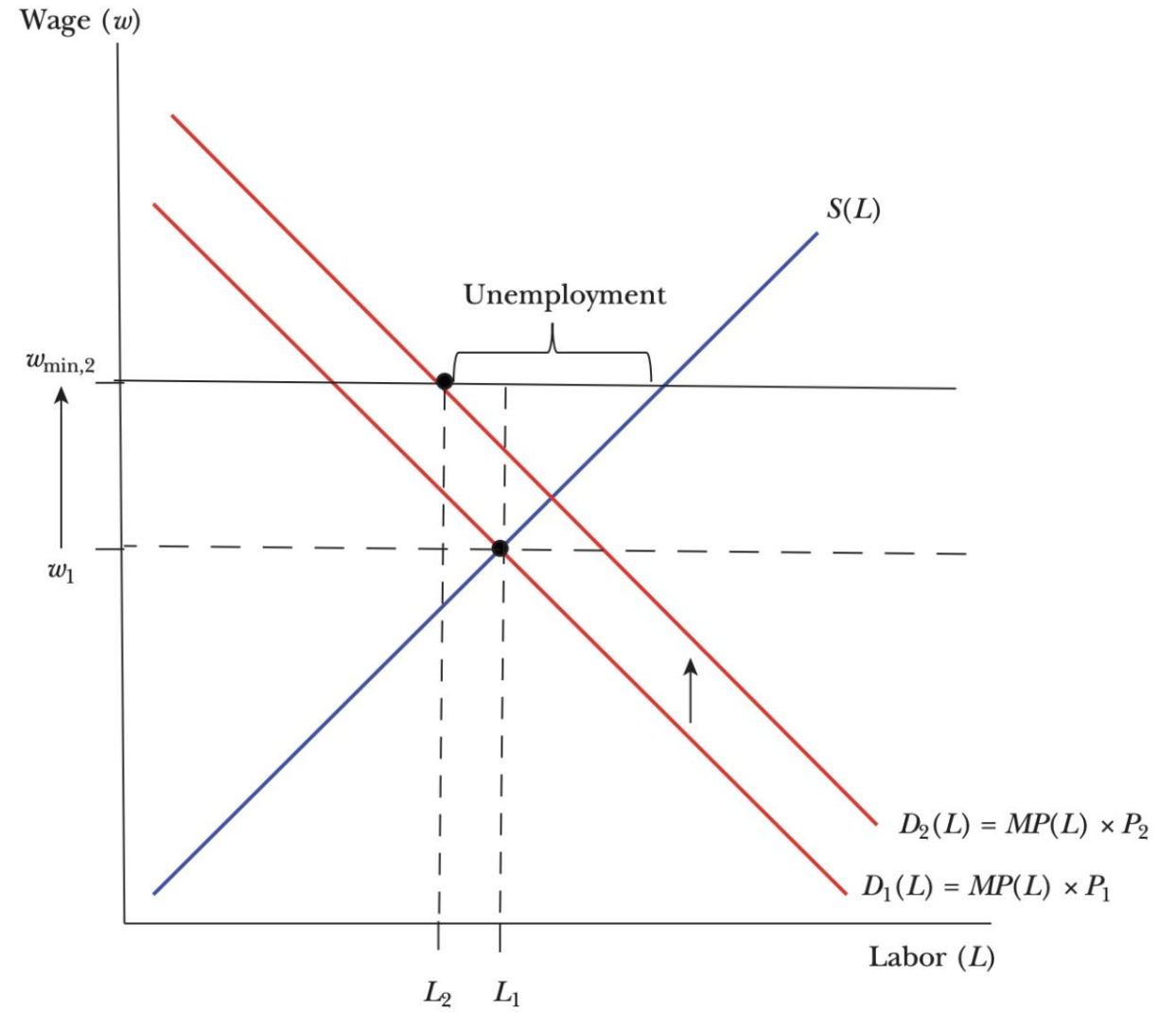
Price Changes

- Marginal Revenue Product of Labor depends on both prices and output
- If firms can adjust price, then this will increase the MRPL for each additional worker
- This will mute the employment effects of minimum wage changes



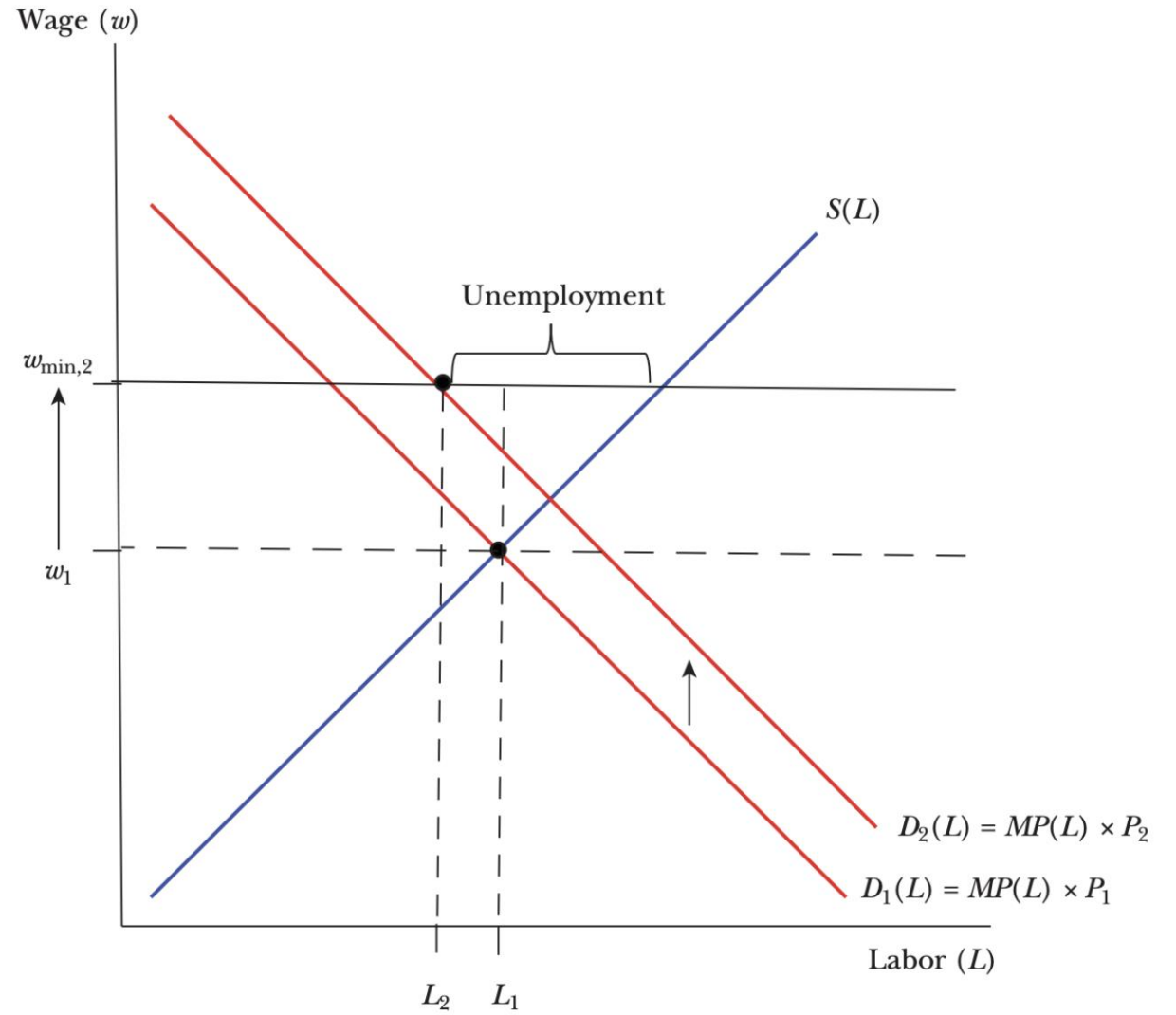
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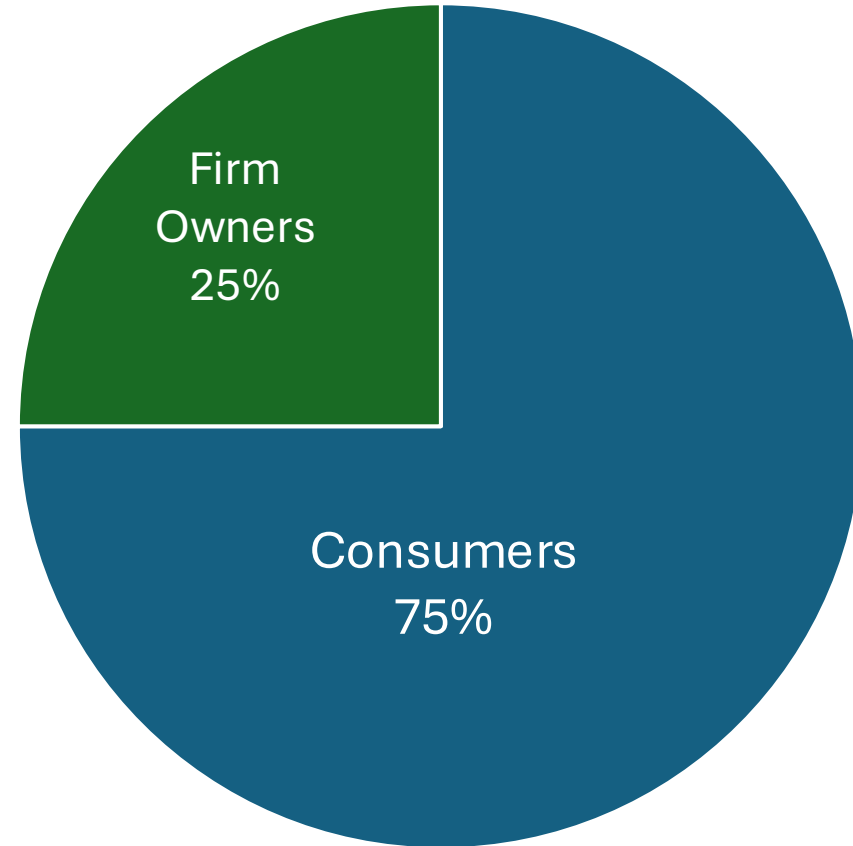
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Price Changes

- Harasztosi and Lindner (2019) provide compelling evidence on this in Hungary
- A large minimum wage change greatly increased cost of labor in Hungary
- They find 75% of increase in cost of labor paid by consumer

Incidence of a Minimum Wage Increase



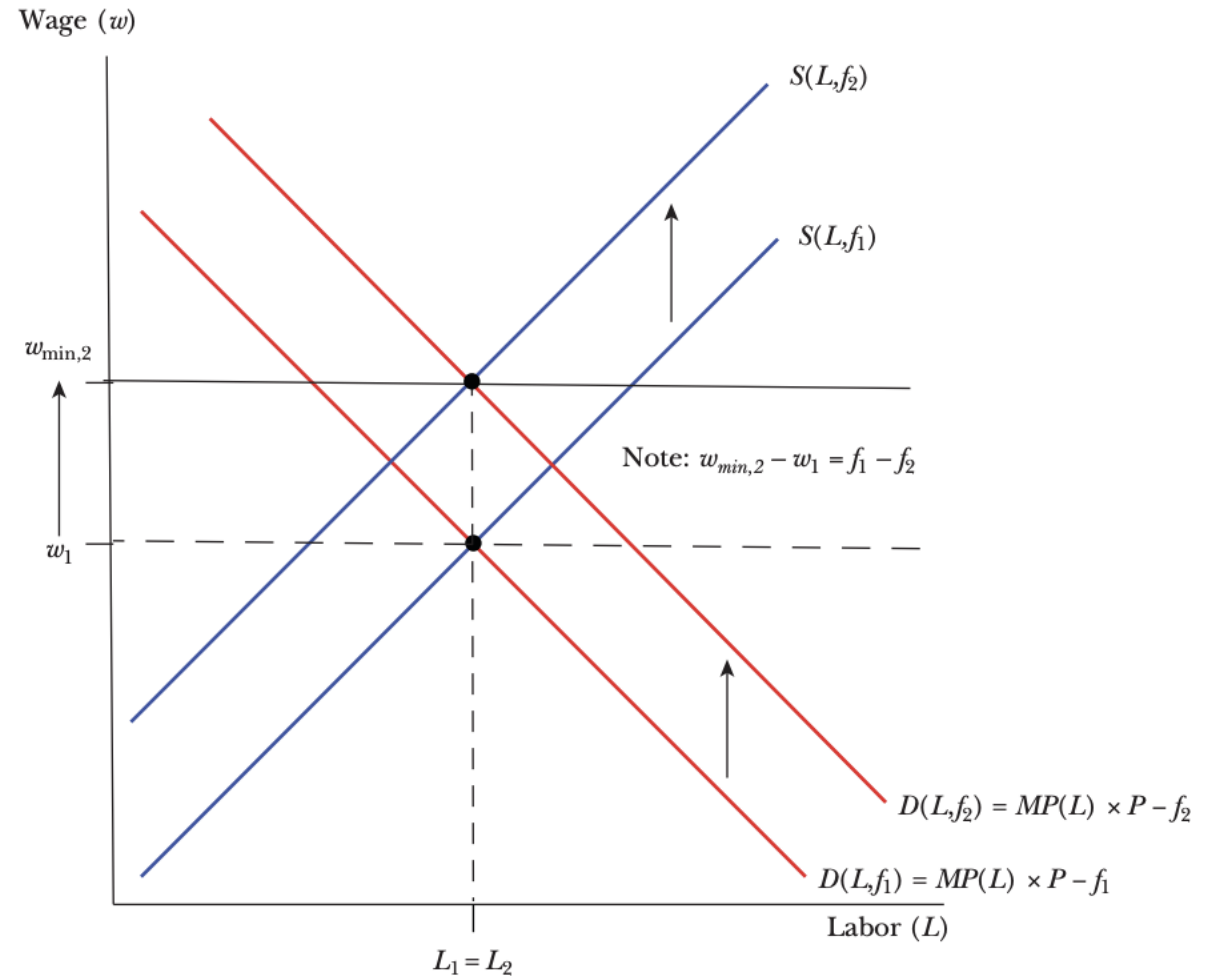


Fringe Benefits

- We have assumed compensation is only due to wages
- But firms also have additional benefits they offer (health care, gym memberships, etc)
- Therefore one way to respond to a minimum wage increase is to decrease compensation in other areas

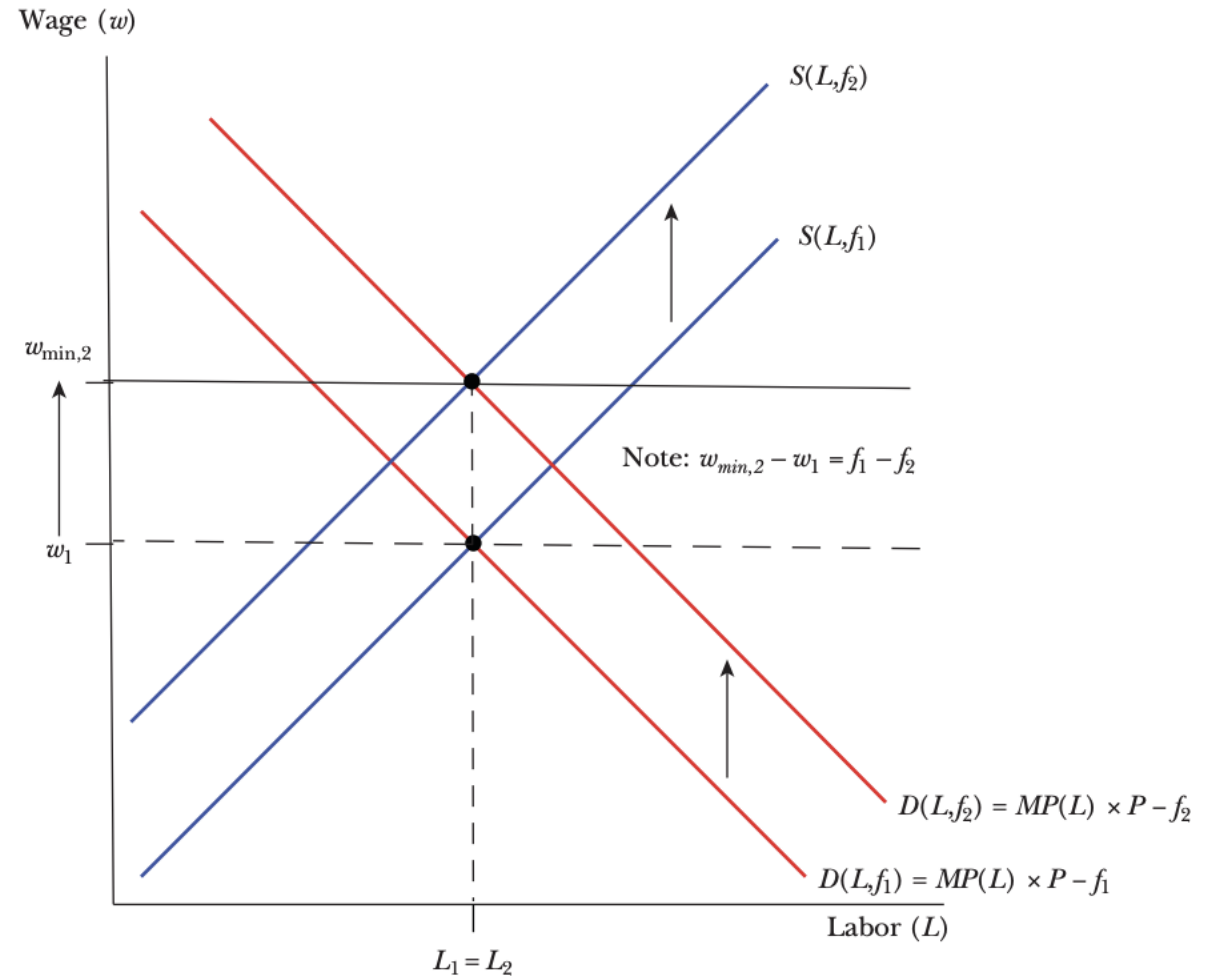
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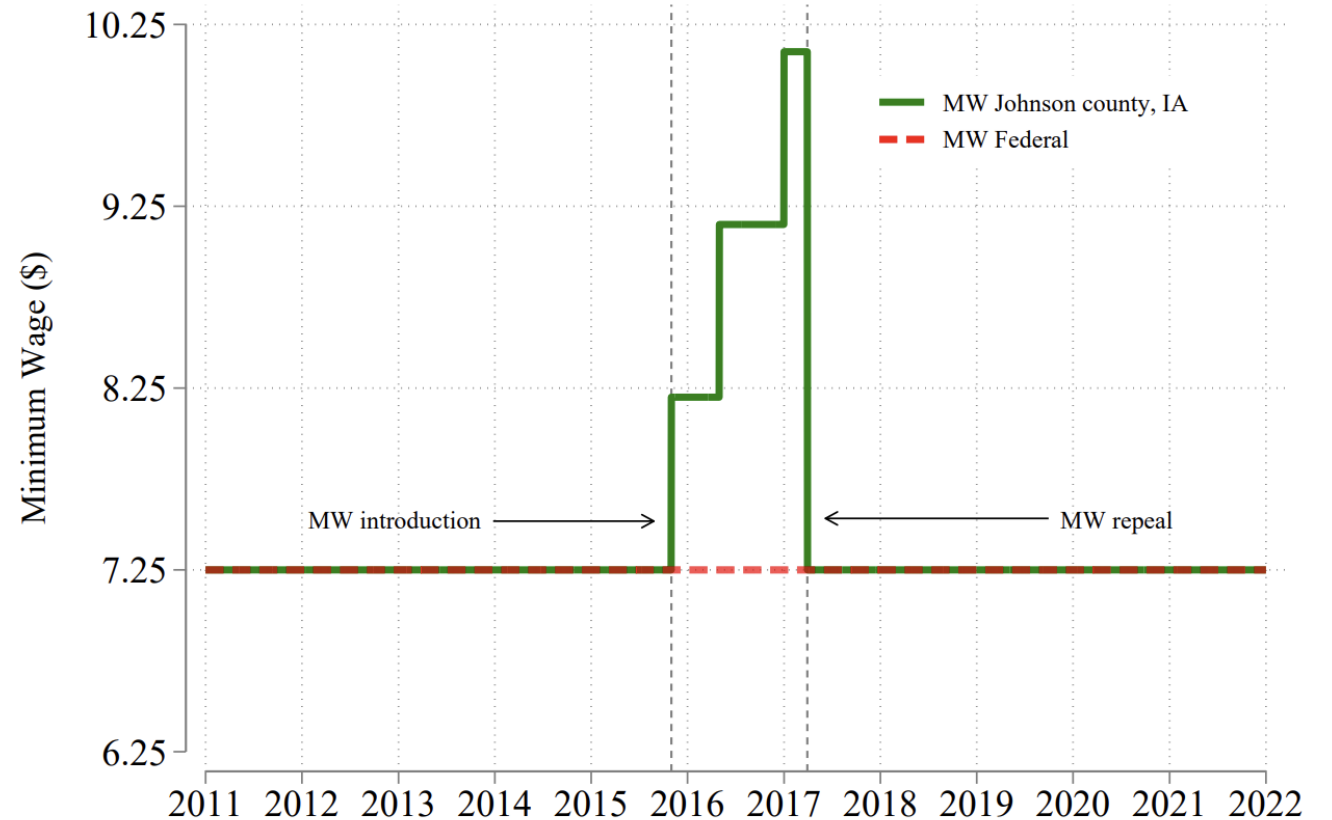
Fringe Benefits

- Clemens, Kahn and Meer (2018) study the impact of minimum wage laws on employer-provided healthcare
- Low-wage occupations about 2 percentage points less likely to receive employer-provided health insurance after min wage increase
- Implies 9% of increase offset by decreased fringe benefits



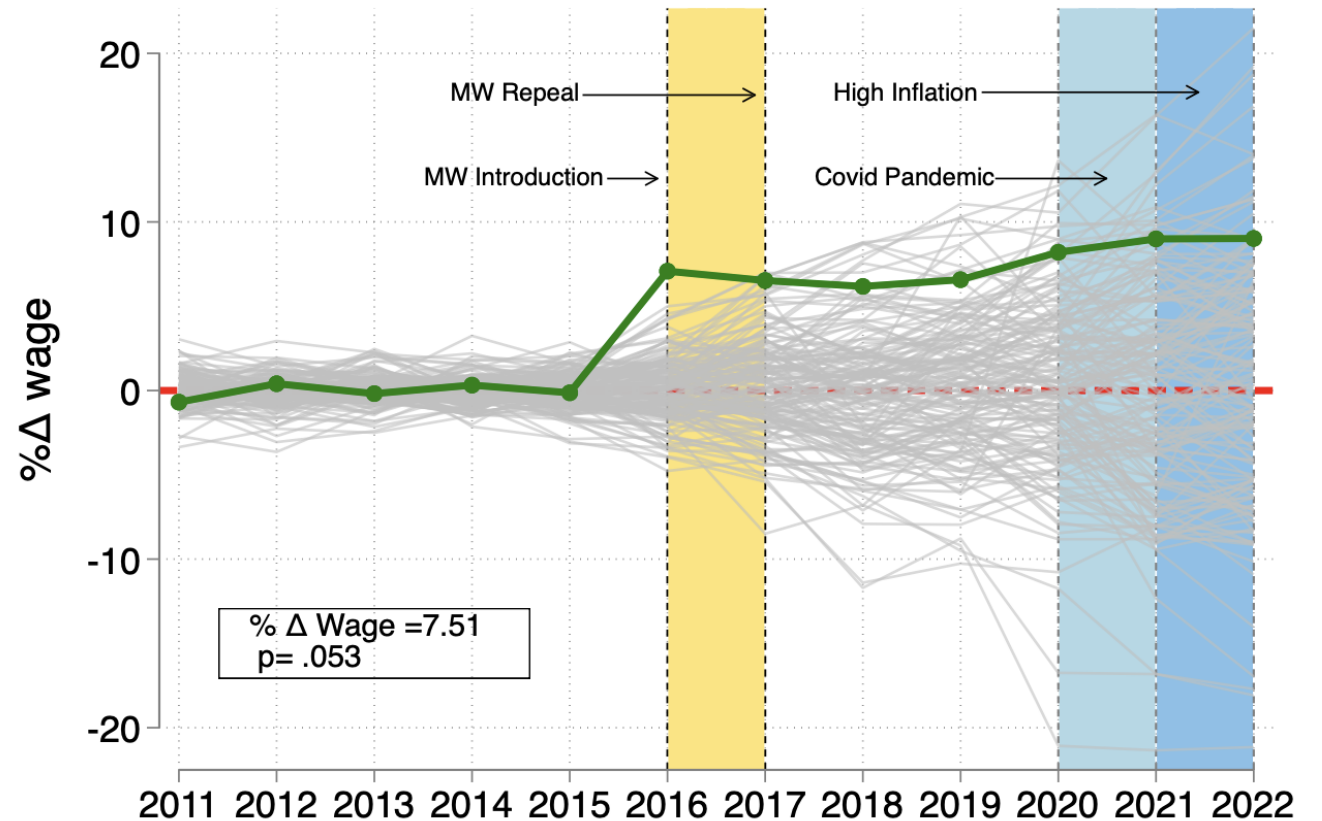
Unexpected Persistence

- In 2016, Johnson county, Iowa passed an unusually large minimum wage increase
- However, the state government made it illegal for counties to have their own minimum wage
- Therefore, the minimum wage at the beginning of 2017 dropped immediately



Unexpected Persistence

- Despite the drop in the minimum wage, the wage increases due to the policy persisted
- This is evidence of wage hysteresis
 - Hysteresis is when a system depends on its own history. The fact that there was a temporary increase in wages implies a persistent increase



Inequality and the minimum wage

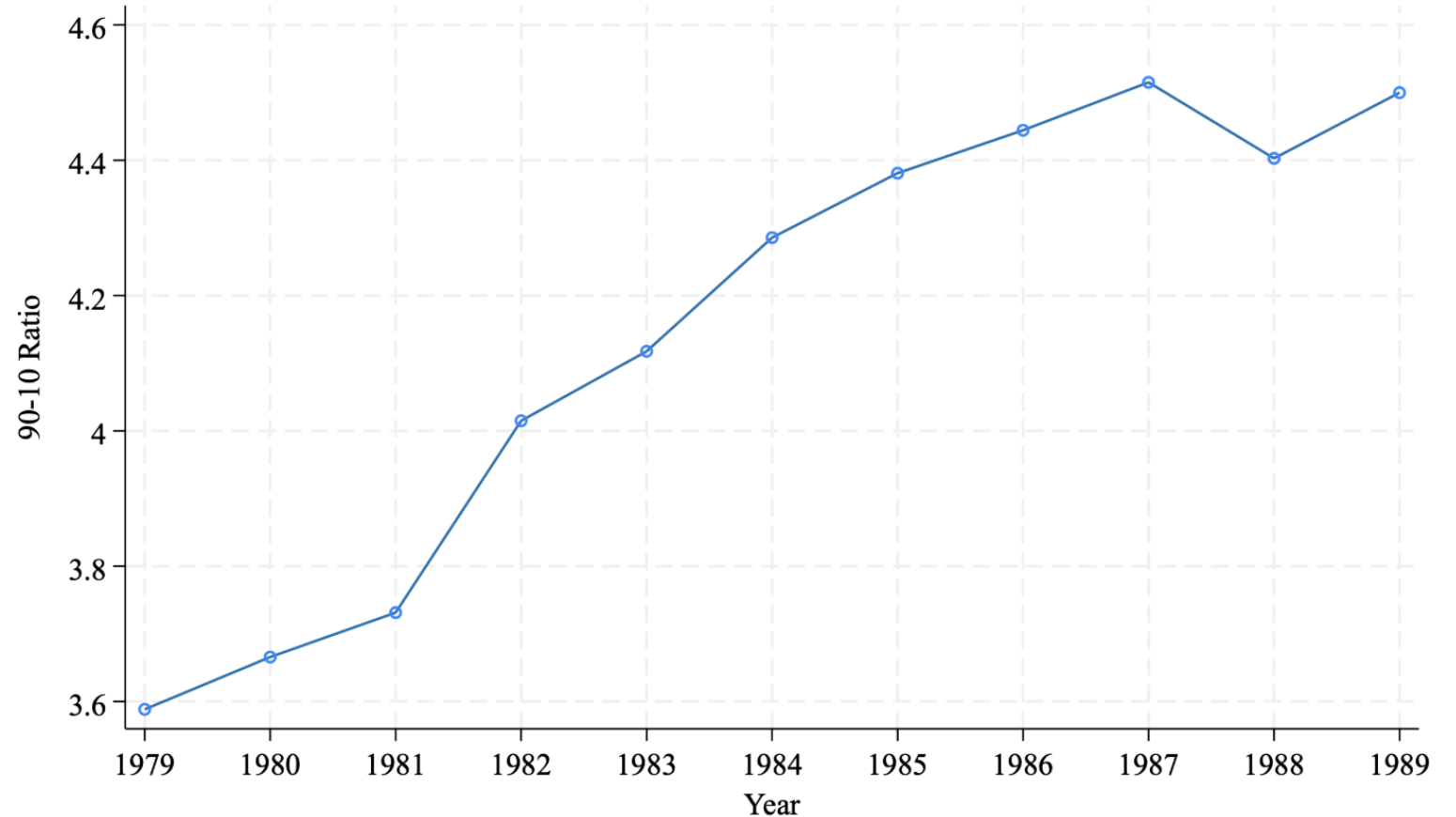
- Let's get back to one of our big questions we opened the class with: the rise of inequality
- We are going to focus now on the period 1979 to 1989, which saw a rapid increase in inequality
- Before getting to the empirics of how the minimum wage impacted inequality, we will introduce an additional measure of inequality

The 90/10 wage ratio

- The 90/10 wage ratio is equal to the 90th percentile of wages divided by the 10th percentile of the wage
 - 90th percentile is the wage you need to be making more than 90 percent of individuals
 - 10th percentile is the wage you need to be making more than 10 percent of individuals
- In 1979, the value of this was 3.6 (i.e. someone at the 90th percentile would make 3.6 times as much money as someone at the 10th percentile)

The 90/10 wage ratio

- Could be driven by two forces
 - At the top, incomes are growing very fast
 - At the bottom, incomes are stagnating



Inequality and the minimum wage

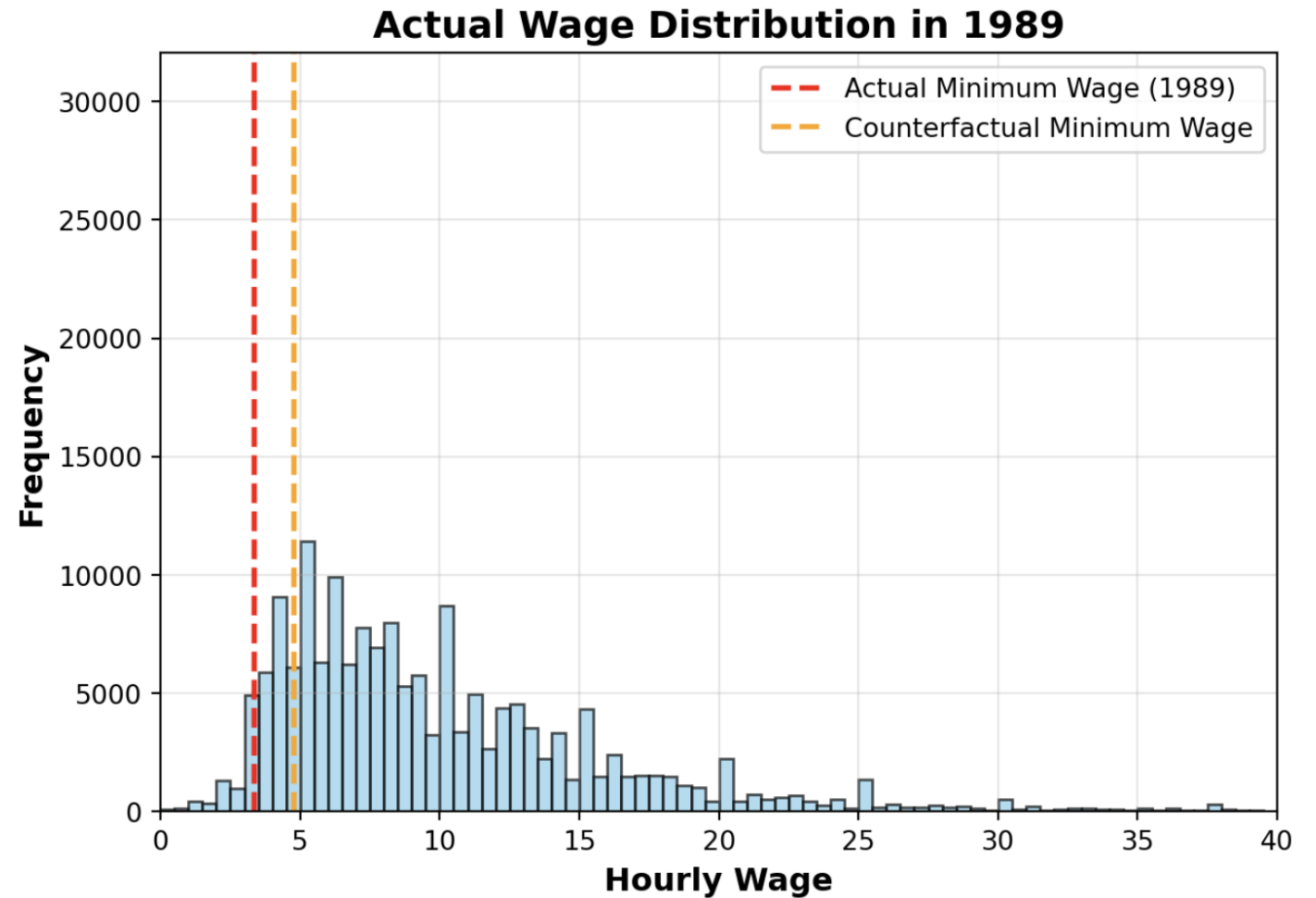
- One reason the 10th percentile might be declining is if the minimum wage is not keeping up with inflation
- Throughout the 1980s, this period of rapid increase in inequality, the federal minimum wage did not change once, remaining \$3.35 an hour for the entire decade
 - The current \$7.25 has been in place since 2010!

Inequality and the minimum wage

- We can use data from the Current Population Survey (CPS) to study this question
- The data we are going to use has additionally been cleaned by Autor, Manning and Smith (2016)

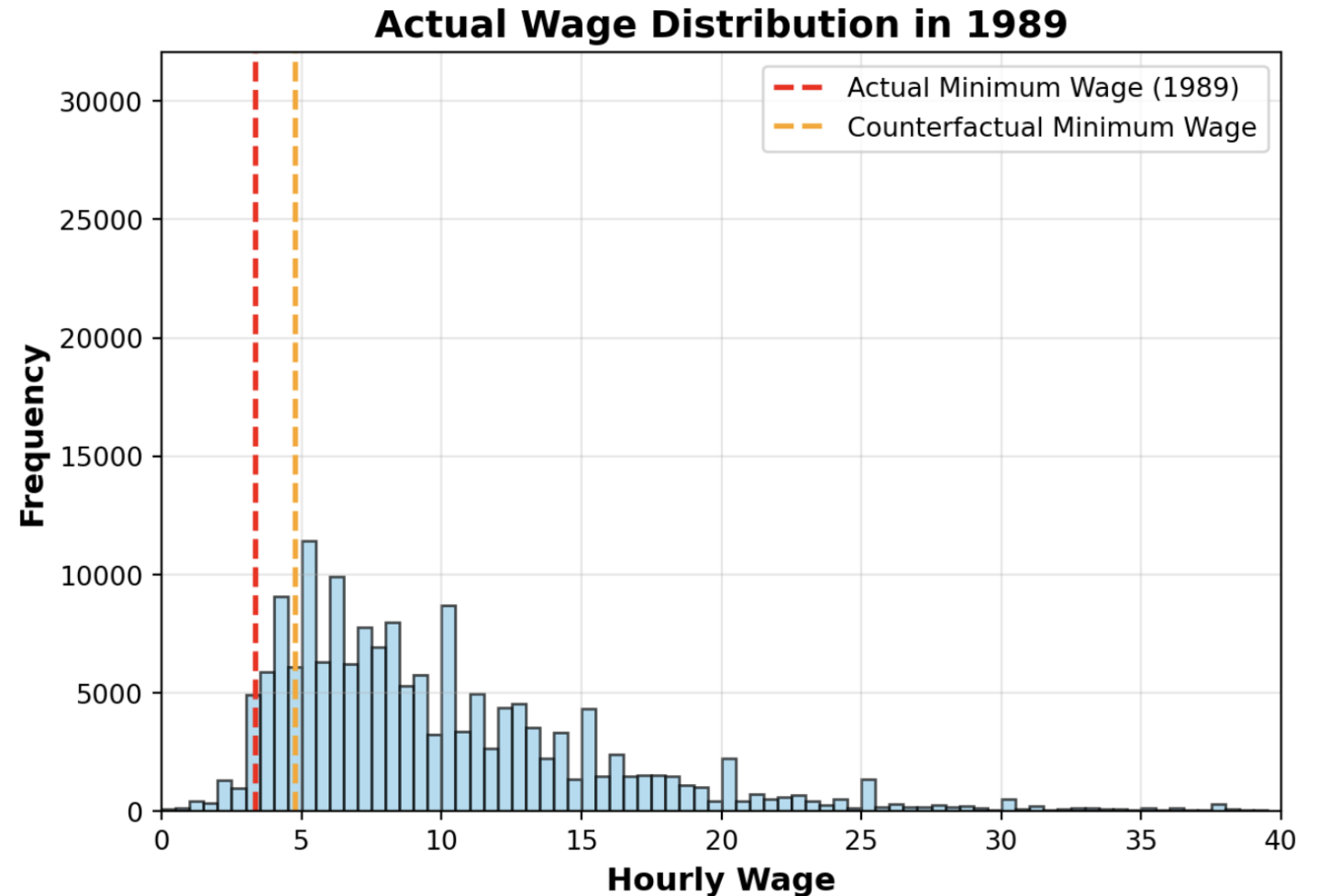
A simple exercise

- The minimum wage in 1979 was 2.90, adjusted for inflation this is 4.77 in 1989
- But the minimum wage in 1989 was only 3.35
- So what would happen to inequality if we moved everyone below 4.77 to earning



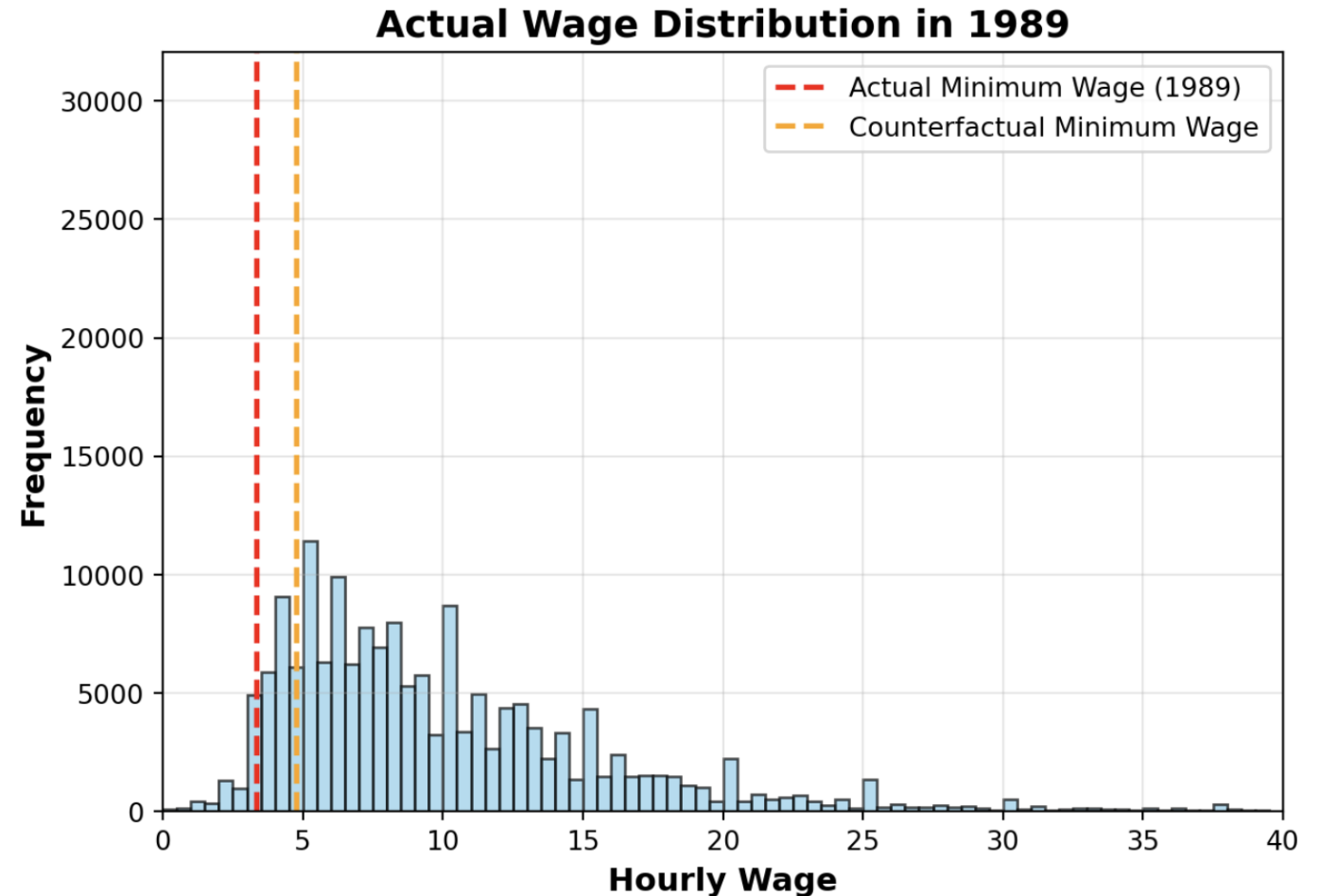
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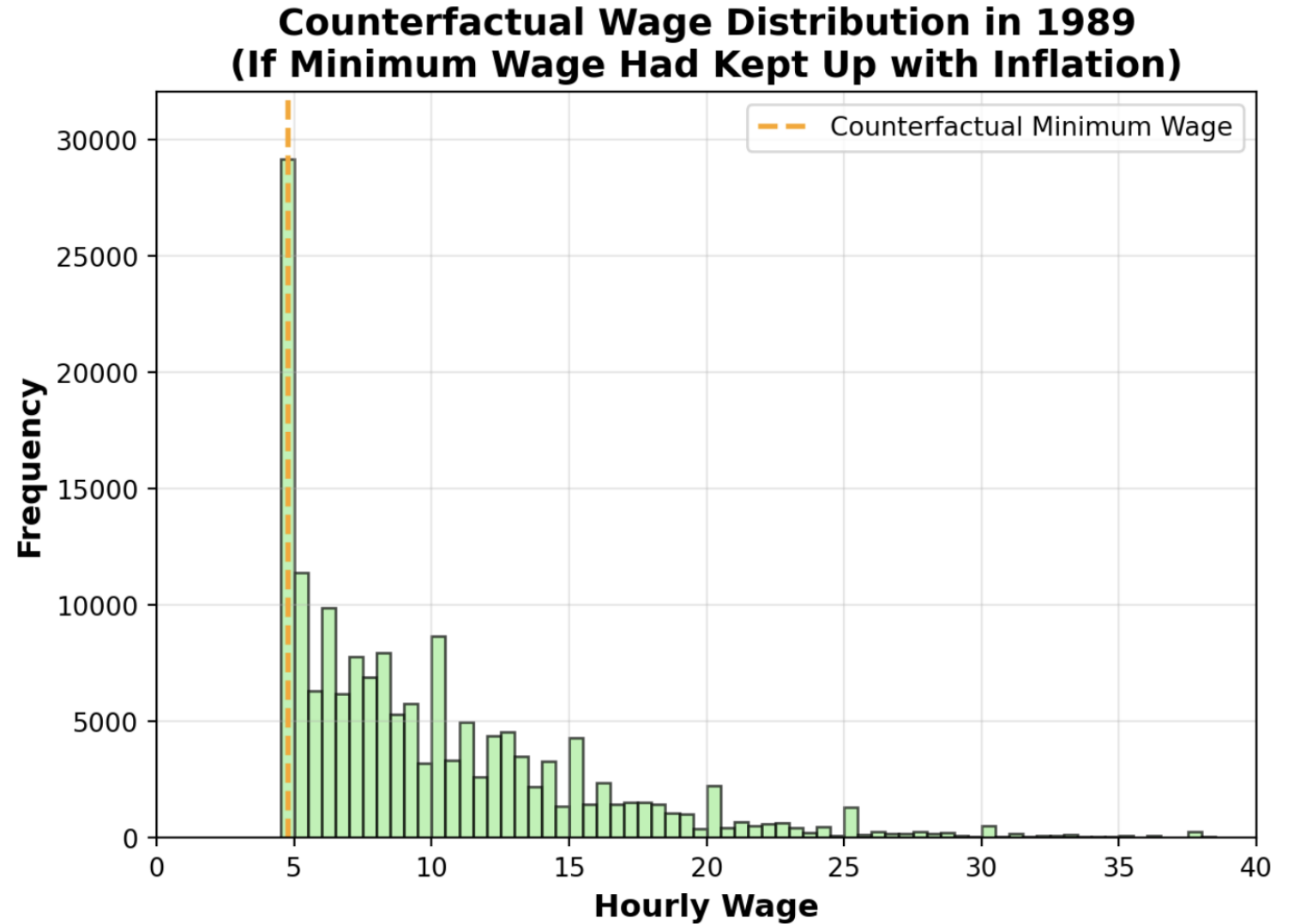
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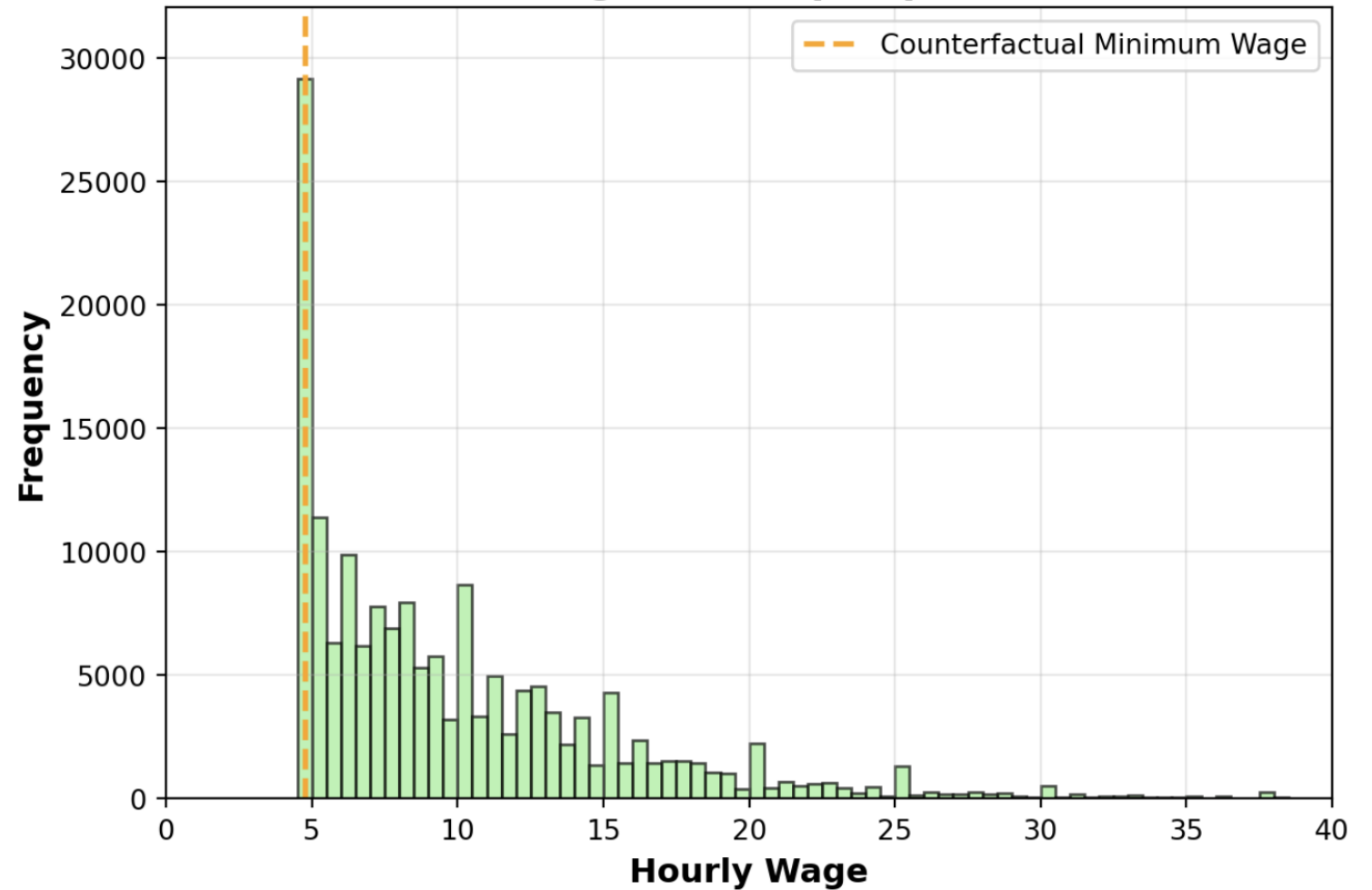
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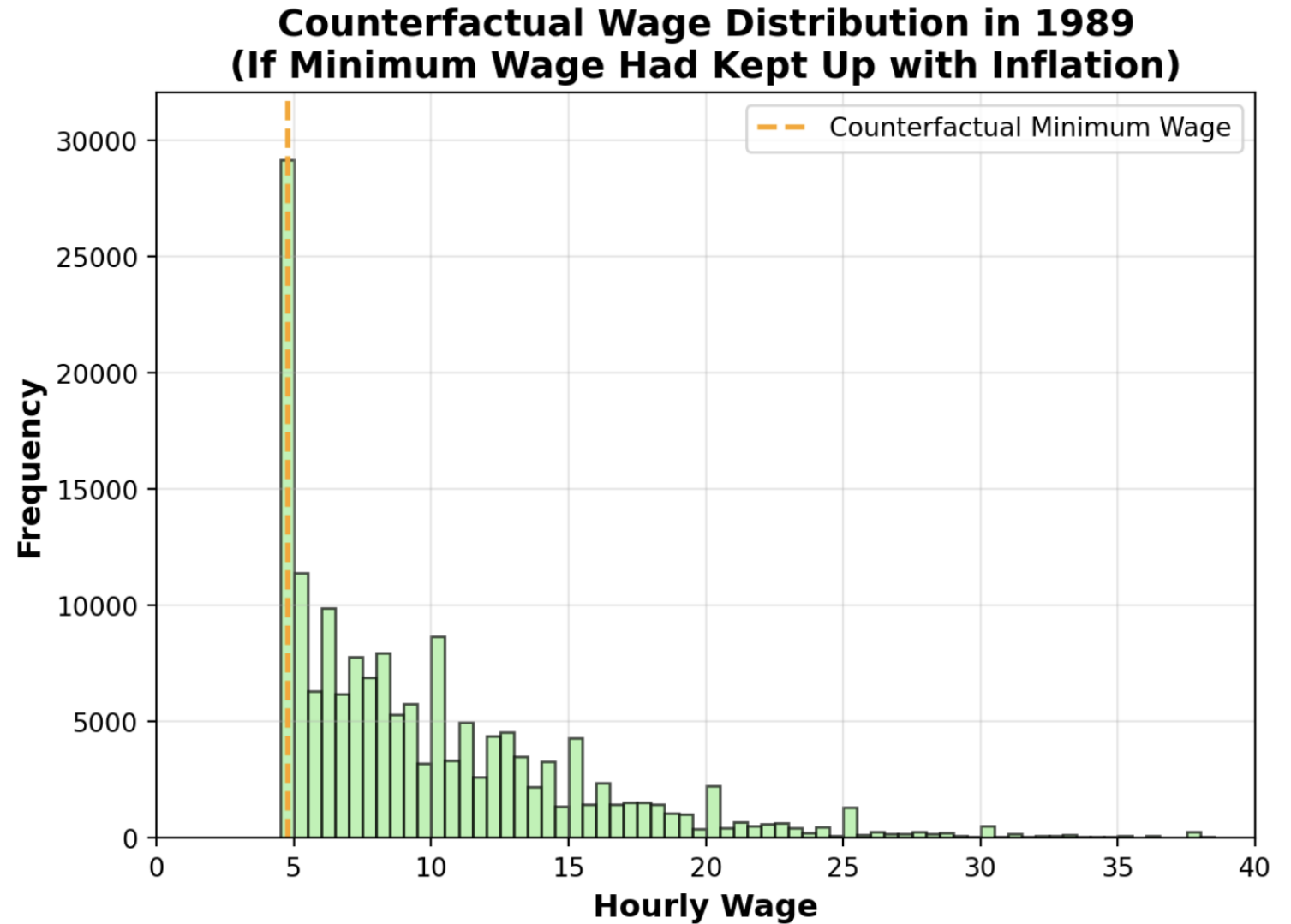
- Let's re-calculate the 90/10 wage ratio with this new distribution
- For this distribution, the 90/10 wage ratio is 3.8, very close to the 90/10 ratio that was in 1979!
- According to this simple exercise, the 78% of the rise in 90/10 ratio can be attributed to the minimum wage

**Counterfactual Wage Distribution in 1989
(If Minimum Wage Had Kept Up with Inflation)**



A simple exercise

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What we ignored

- Non-compliance (you could see some individuals were below the legal minimum wage) but our exercise assumes everyone will be put above
- No employment effects. Although the empirical literature finds small negative effects, a small amount of negative employment effects could impact this number
- No spillovers (there is work finding if you raise wages for the lowest-paid workers, slightly-higher workers will also receive raises)

A More Careful Analysis

- DiNardo Fortin and Lemieux (1996) is a seminal study that performs a careful decomposition of how much of the rise in inequality can be attributed to the minimum wage
 - They attribute 40-65 percent to the minimum wage (recent evidence finds that looking at the entire period 1979-2012, minimum wage can explain about 30%)
- It's important to take a step back and realize how surprising this is. The minimum wage is only one policy. The 1980s was a period of rapid technological change, globalization, etc.
- We will see that rapid technological change is indeed an extremely important factor in the wage structure, but it is surprising how much we can explain just considering changes in the real value of the minimum wage

Conclusion and Next Steps

- The results in the minimum wage literature are surprising when viewed through the lens of perfect competition
 - In a subset of studies, researchers have even found positive impacts
- In the next set part of the class we will consider an alternative way to look at the labor market: monopsony